

Golden Science

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Abstract

Currently in the study of natural history, and how it integrates with and perhaps even equates to Earth science, there are two generally accepted methods of gaining new information. The first, and often most telling, is to physically dig up a creature or scene from the underground and let it touch the sun of knowledge for the first time since countless many eons ago. Second then, might be genetic analysis, harnessing the incredible power of genetics present in every one of us to trace back the clock and ask the question of when certain modern clades had their origin. Now, though, with the power of mathematics and the mediocrity principle at our side, it is possible to create a third method: Aurusperspective Reasoning, as I call it. Aurusperspective Reasoning is a certain very specific type of reasoning which takes its inspiration from some concrete known, and the logic of the mediocrity principle, to generate new information that would be otherwise extremely difficult to uncover, if not outright inaccessible using other methods. In the following pages, I have listed three main laws of Aurusperspective Reasoning, along with a few starter cases of where they may be used to reveal the past of the Earth in a brand new way.

1 Introduction

This paper is, first and foremost, an extension of the Rare Earth Hypothesis [1], which fundamentally states that intelligent life is a rare occurrence in the universe, explaining why, among other things, we have yet to detect extraterrestrial civilizations in the outside cosmos. The evidence for this hypothesis is notoriously difficult to process against its various counterarguments, but it nonetheless seems to be the dominant player in whatever new studies and thought experiments arise throughout the scientific community. It is as plain as it is straightforward, and perhaps terrifying for it. But without a general purpose, or a landmark so to speak throughout its nature, it is difficult to comprehend such a reality, because, for instance, how can one call intelligent life rare without a formal definition for intelligent life? In this paper,

such a definition will be proposed. Or rather, if our species qualifies as intelligent life, then a definition will be proposed for what the natural tendency of such life is, using pure statistics. See the section entitled “The Golden Point” for the explanation of this. The fundamental tendency and cosmic-scale classification of life may be more tangible than first meets the eye. Once such a landmark is considered, now the entire extent of life, meaning its history, development, and state in the universe, may be identified with patterns and laws by which it abides.

The backbone insight behind all of this is to use the mediocrity principle to our advantage, and consider what such a principle must tell us about ourselves and our history on this planet. In short, we may be quite capable of setting up arguments which either must be true, or violate the mediocrity principle outright. And even where the mediocrity principle is potentially violated, bringing such states to the baseline of observation is essentially improbable anyway, such that, so long as our supposed observations are accurate in these cases, the statistical chance of certain logical conclusions based upon this being in a state of falsehood is intriguingly unlikely. For instance, anywhere where a Great Filter [2] is present, the mediocrity principle may yield at least a few insights about the world without that happening, and even the world of before the Filter. At its best, this logical system may even uncloak ghost Great Filters which would not be otherwise accessible. For such Great Filters, the likelihood of them being so is overwhelming.

But this is only the beginning. Combining both of these disciplines – the uncovering of hard-to-find natural history, with the great landmark of life itself, points directly to one fascinating result: our generation, in existing at all, occupies a wild point of importance in the cosmos, itself on par with the question of why the entropy of the universe is so low to begin with. It seems as though our very existence is at the pinnacle of when the old biological universe transitions into the new biological universe so to speak, and the answer to why is at this point strikingly unknown. And from this vantage point, it potentially becomes possible to decode some of the unknowns in our reality, and ask what function of our universe

truly puts what we see before us.

Aristotle once said that the rain falls for the sake of the grass, and not out of pure necessity. Later in history, this was rebutted, arguing that the rain simply fell for no particular reason, and that the grass evolved to take advantage of this rainfall. At the end of this paper, a third argument will be made, which states that they are both, in their own ways, correct, and that the real answer is that more of relativity. In order to give a subject purpose, (or lack thereof), it is required to describe it in terms of some other subject. That is, with respect to the background coldness¹ of the cosmos, yes, rain is in fact meaningless and does not serve any particular purpose. But relative to human observation? Relative to that, and assuming the Golden Principle which is to be suggested later on in these pages is indeed valid, the real answer would be more along the lines of this: *The rain optimized to enable the Golden Point, and some particular grass evolved to take advantage of this happening.*

2 Invalidation of the Cosmological Mediocrity Principle

Let us start this discussion by dethroning the currently leading hypothesis for the mediocrity principle's application to the cosmos at large.

The mediocrity principle is something of a beautiful tool for predicting one's placement in a set given the simple attribute that this placement is random. For instance, if there is some mystical bag that carried a billion and one colored balls – a billion of them all being either red or blue, and the other one occupying the other color, then if you could reach your hand into this bag and pull out a random ball, say a blue one, and now you know with near certainty that the billion balls were all blue, because otherwise

¹This “background coldness” is a stand-in for the concept to be proposed later on in the paper, called “Outlook Probability”, which is meant to be indiscriminate to any specification of happening, thus making it broad and objective. The Copernican Principle, for instance, is valid with respect to Outlook Probability. But relative to other things, and the Copernican Principle has the potential to break down as a valid function of reality.

you have just undergone a one-in-a-billion rarity of happening.

This same tool is often applied to the Fermi Paradox, in a seemingly clever approximation known as “The Cosmological Mediocrity Principle”. However, according to the logic of this paper, the cosmological mediocrity principle is mistaken in its fundamental conclusions. The simple reason for this is due to poor assumptions and lack of awareness regarding said assumptions. Specifically, this principle assumes that planets and life are fundamentally linked in some way, however this is clearly uncertain and even arbitrary. The cosmological mediocrity principle essentially suggests that since life exists on Earth, then it should exist on Earth-like planets universe-wide. But even arguing this is heavily based on vague information. Namely, how does one define, “Earth-like planet?” The word planet itself is a human construct. Planets are not like atoms – some fundamental normal of the universe – rather it is a word first designed by the ancient Greeks to track the motion of dots across the celestial sphere. Later, when the International Astronomical Union gave their definition, they were just picking more human-centered defining attributes which are based in objectivity no more than the word croissant or bicycle. We could easily flip the definition of planets on its head, and objectively we would have no reason not to. If we did such a thing with atoms, we would be clashing with the concepts of basic fundamental chemistry, and the simple binary-like properties of things would fall apart. Sure, we could redefine atoms as, “the old definition of atoms and or a bicycle”, however in doing that, we would still destroy the clean mathematical-like definition of things that is rooted in our current definition of the atom. With a planet, there would be no such issue. Take, for instance, brown dwarfs. Are they planets? Stars? In a way they are somewhere in the middle, which makes evident the flaws in our black-and-white thinking with planets as are described in the IAU's rules. So the arbitrary factor of the planet definition severely weakens the argument of the cosmological mediocrity principle just by itself. Worse yet is the “Earth-like” addition to the statement. What makes something “Earth-like”? Is it a roughly Earth-like time dilation? Is it a region with a roughly Earth-

like temperature? Is it a gravitationally bound mass above a certain radius which contains more than one part per trillion water? Here is a simple *reductio ad absurdum* thought experiment for the cosmological mediocrity principle: let us imagine that tomorrow the IAU redefines a planet as a cubic-centimeter sized region of space which has an average temperature below the Planck temperature. Perhaps they also argue that to be “Earth-like”, a planet must have an average temperature which is actually no more than half as much as the Planck temperature. So far no further assumptions have been made than the currently decided definition of “Earth-like planet”. The arbitrary factor is thus equivalent between the two. Continuing on, with this newly defined definition of Earth-like planet, the cosmological mediocrity principle now determines that since life is known to exist in all Earth-like planets that are directly observable to a person, then it must be common for life to exist in the majority of Earth-like planets universe-wide. The conclusion is that every little region of space is filled with life, be it in the vast coldness between galaxies, or in the hearts of neutron stars or supernovae. It may even be calculated that the universe is mostly life by weight, if one were to pick just the right life-sample in their Earthly observational reference view. The same amount of assumptions have been made, and yet this yields a completely absurd result. The point of all of this is that while its conclusion may or may not still be valid, the actual logical process of the cosmological mediocrity principle is completely baseless.

3 Finding Prerequisites With the Mediocrity Principle

Now comes time to state the basic derivations of Auruspexive Reasoning. More rigorous formulations are still to come, but a simple starting derivation is nonetheless very much accessible using the mediocrity principle alone. Without further ado, let us begin.

One famous application of the mediocrity principle is regarding one’s birth index in all of human history,

which hints perhaps at how long into the future humanity might live to [3]. This paper seeks to generalize this concept even further, and consider alternate dimensions. Sure, we can manipulate the three dimensions of space around us, and to a lesser extent the temporal one, but calling this simplistic concept the full scope of the mediocrity principle’s power is somewhat arbitrary. If we imagined that the universe was infinite in extent, stretching as far as can be imagined in all directions, and having roughly similar physical principles all throughout, then, statistically, one could expect to find repeats of humanity and the Earth now and again. Thus, by the mediocrity principle, rather than fundamentally visualizing oneself as a random human birth on Earth, it would be better to imagine oneself as a random human birth in the infinite cosmos. This seems rather intimidating of a mathematical idea, but luckily with just the right manipulation, it can be wrapped up in its entirety to a single spatial location, varying only with time. Specifically, by use of timeline splits. If there was a far-off human bearing planet on the other side of the observable universe, then we can simply imagine generating the set of happenings which generated this distant planet by changing just the right events in Earth’s past, such that we mimic this far-off planet. If we do this thoroughly enough, we will create an exact duplicate of it in place of the Earth. Thus, just by imagining the set of every possible timeline split in Earth’s past, and where these alternate realities lead, we have generated the equivalent set of all the life in the entire infinite-stretching universe, to the exact same proportions and mathematical structures at that. An intuitive way to visualize how the mathematical trends even out is regarding the way that less likely timelines will occupy less of a number of the total set of all timelines, and by the exact same process, such generated worlds from those timelines out in the vast depths of space will be proportionally more rare than more common planets/timelines by the same circumstances.

As for a quick mathematical analysis which shows this to be true, it first must be recognized that the timeline split itself is actually a constant phenomenon, not dependent on variables of difference. One can visualize it all as the timeline split itself, fol-

lowed by the set of every possible happening which can follow that split in time, themselves being categorized into two buckets depending on whether some attribute was made present or absent at the moment of the split. Those buckets are then labeled “Timeline 1” and “Timeline 2”, allowing the full picture to be built. Now, because this picture does not vary across any type of external change, it can be reasoned that the total number of humans present in the sets of Timeline 1 and Timeline 2 are constant, so therefore the ratio between these people, some r , and thus the relative probability of birth ratios between these two Timelines, again that r , is constant in the picture of the timeline split. This means that even given the tiny number of, say f , the proportion of worlds out of all possible universal worlds which reach the event of this exact timeline split itself, even though this number affects the probability of being born in Timeline 1 vs Timeline 2 relative to some earlier time, still their ratio of r must be constant. For instance, if the probability of birth on Timeline 2, relative to the timeline split, is some x , then the probability of birth on Timeline 1 is rx , and so relative to some time beforehand which gives the timeline split a probability of f , the probability of birth on Timeline 2 is xf , and the probability of birth on Timeline 1 is $rx f$. Running the ratio $\frac{rx f}{x f}$ of course just yields the constant r . Hence, for any timeline split that can be imagined in Earth’s history, the exact ratio of worlds in the infinite cosmos which contain as opposed to lack some attribute of difference defined by the timeline split is still just that same r . This, exactly, is what drives the logic behind this model. And of course if any one timeline split can be considered, then so too can every potential timeline split, therefore yielding this concept. (Note that versions of this logic still hold true whether or not the Cosmological Principle is valid – the fundamental properties of reality at the location of the Earth can simply be added characteristics to the timeline split itself, and then the remainder of things work as before).

Thus, arguably even more fundamental of a use of the mediocrity principle than considering the set of all people in Earth’s history, is considering the set of all people in every possible alternate reality that could have triggered in Earth’s history. And doing

just this yields some fascinating results.

To get the general idea of such a concept, a great place to start is with the Great Filters [2] of the Fermi Paradox, or in other words, the events and circumstances which had to trigger perfectly right in order to generate humans. Take, for instance, the death of the dinosaurs. What are some logical conclusions that can follow given the argument that the death of the dinosaurs was a Great Filter? In order to solve such a thing, a key element is understanding some of the probabilities surrounding the death of the dinosaurs. The universe itself is a very dangerous place, so the death of the dinosaurs in and of itself is perhaps something of a likely outcome, especially given the vast spans of time associated with Earth’s geological history. In most of these cases, such a thing is part of a worldwide obliteration of life, due to a whole range of disasters such as supernovae, asteroid collisions, and the rest of the ways outer space is eager to kill us at any given moment. What is probabilistically harder is the death of dinosaurs corresponding to the survival of human ancestors. Now a simple sterilization of a planet does not suffice, as human ancestors die too, and so a far finer and more precise extinction is required. And the direct consequence of this is that the probability of such an extinction drops substantially. One lead, however, regarding it, is the makeup of oceanic water on the surface of the Earth, and how such things pertain to the event that is the death of the dinosaurs. A deep ocean asteroid strike would have cushioned the impact energy too much to kill the dinosaurs [4], and a land strike similarly would not have kicked up the same tier of devastation. Either way, the dinosaurs would likely live². Thus, if it is taken as true that a shallow oceanic strike would

²There is actual evidence of land craters of roughly the same size which never did cause mass extinctions [5], so such craters just further the reality that land strikes subtract from the devastation – a concept which would ensure the dinosaurs did not die. Also, even if you are not one to prescribe to the depth of waters as a factor idea, the way that this leads into the sulfur concepts is nonetheless intriguing. This means that the sulfur deposits present at the blast site may very well be the main contributor to the flukey nature of this mass extinction more so than the water depth, but either way, the main point is unchanged: this was a very unlikely event to actually succeed, all things considered [6].

be the only means of destroying the dinosaurs but not the mammals, then it becomes a requirement for the asteroid to strike shallow oceanic water. Because approximately 5% of the Earth's surface is shallow oceanic water, this means there is no more than a 5% chance of the Great Filter occurring.

Taking this into account, if the probability of humans evolving given that the dinosaurs are dead is some value a , and the probability of humans evolving given the definite survival of the dinosaurs is b , it follows that the total probability of humans evolving is $5\%a + 95\%b$, which is simply a weighted average of these two probabilities as determined by the probability of the Great Filter occurring in the first place. As a generalization, where the Great Filter has a probability of occurrence of p , and the odds of humans existing in its success and failure are a and b respectively, it follows that the total likelihood of humans existing is $pa + (1 - p)b$. Where the Great Filter's success and failure represent a timeline split, this value tells you the "mass" on the two timelines of total humans. (Mass here being defined as some proportional value to the weight that would be achieved if all humans in a specific set of timelines were put into some grand scale. Note that this also has to be proportional to the probability of the timeline being reached by prior splits and human likelihoods, at least if we assume there are roughly equal number of humans to a timeline, which most likely is a very valid assumption given the averaging process of higher human numbers in some timelines balanced with lower numbers in others – not to mention other near constants like carrying capacity and land area). Peering into this mass, what proportion of it exists on the timeline where the Great Filter is successful? Well, that would have to be $\frac{pa}{pa + (1-p)b}$, because such a thing measures the ratio of the mass of humans in the successful Great Filter timeline to the mass of humans in total. In the case of the death of the dinosaurs, this looks like $\frac{5\%a}{5\%a + 95\%b}$. This value now measures, at least relative to just before the end of the Cretaceous period, what proportion of all humans in all timelines branching from that time will be born in timelines where the dinosaurs have been wiped out. The rest of the humans would have been born in time-

lines where the dinosaurs survive. This is the moment where the mediocrity principle becomes vital. Now that the proportions of humans are already well known, the key is to consider the likelihood of being born in one set of timelines as opposed to another. If there is truly a random chance among all humans in all timelines, then in order for us to be born in the dinosaur extinction timelines, by the mediocrity principle we would expect it to contain the majority of humans. Otherwise we would be born in the minority, which is less likely. From this, at least according to the logic of the mediocrity principle, we can write the following inequality: $\frac{5\%a}{5\%a + 95\%b} > \frac{1}{2}$. Taking this into account, algebra yields $5\%a > 95\%b$. Then, lastly, by manipulation to get the ratio from b to a , it can be determined that $\frac{b}{a} < 5.3\%$, or equivalently, $\frac{b}{a} < \frac{1}{19}$, (which can be additionally expressed as $a > 19b$). This is a fascinating result, because while it tells us no information about the individual probabilities of humans evolving in specific timelines, it tells us that the net probability of humans coming into being in the timeline where dinosaurs live on cannot be any greater than nineteen times less than the probability of humans in the timeline where dinosaurs are officially dead. It is technically possible for this condition to be broken, however such a conclusion would clash with the mediocrity principle if it were to happen, and thus is likely not going to occur. This is one simple means of using the mediocrity principle to determine information about alternate timelines, best used for fluke events like the death of the dinosaurs, which in order to occupy a large number of humans must have to also be vitally important to the evolution of humans by this mediocrity principle logic, specifically through the generalized formula $\frac{b}{a} < \frac{p}{1-p}$.

Note that it would be bad mathematical form to assume the mediocrity principle can be applied to every situation – truly the best outlook on this is to imagine one's birth as random in the set of all humans, and leave it at that. For one single human, the likelihood of occupying the majority of human-mass is good, however the moment more humans are added, the statistical models skew, and no longer is this clever trick as valid for all of them. For multiple humans to all occupy the majority of timelines is actually not necessarily a good probability, because of

that heightened chance that at least one is to spawn in the minority of humans across the timelines. For infinite cases – the equivalent of stating that this form of the mediocrity principle can be used for all cases of timeline splits – and the probability of every situation falling in the majority tends to zero. Hence it is proven to be a bad idea that all timeline splits can follow this majority principle, however individual ones are no less expected to do so anyway. Consequently, when it comes to wild flukes among the history of the Earth, while individually they are each expected to be highly vital to human existence, there certainly are still ones which are useless to the prospect of human evolution in one way or another, however they may be extremely difficult if not impossible to find.

Another great use of this mediocrity model is regarding splits which are dependent on future happenings to satisfy their human evolution. Take, for instance, the evolution of dinosaurs in the first place. This time, rather than declaring the death of dinosaurs as a fluke alone, it shall be declared as a full Great Filter, a kind of subclass of a fluke. The additional characteristic that differentiates a Great Filter from a fluke is the added necessity of being a vast Prerequisite to human existence. If a Prerequisite fails, the probability of humans coming into being is an extremely small number, nearly indistinguishable from zero. In the additive world, through $pa + (1-p)b$, if say p is the probability of some Prerequisite being satisfied, then $b \approx 0$, and so $\frac{pa}{pa + (1-p)b} \approx 1$. While it will never be exactly one, it will be close enough to weed out human life in the failure timeline thoroughly enough that if one could look over all the planets which have humans in them, they could scroll past thousands upon thousands of worlds, and never once find one from the failure timelines. Perhaps thousands is an understatement for truly major Prerequisites. From here, if a Prerequisite is also a Fluke, it becomes a kind of Great Filter. It has already been suggested by the scientific community that perhaps the death of the dinosaurs was a Great Filter, and so what follows is a mathematical analysis of what such a reality would mean for the evolution of dinosaurs in the first place. Now let us imagine a timeline split during the Triassic. In one timeline, dinosaurs evolve

from more primitive archosaurs, and in another, they never come into being in the first place. One might consider this a favorable situation, because if the dinosaurs never evolved, then there would never be the need to wipe them out for the purposes of the later Great Filter, thus dodging that Fluke in full. But if this were true, and the dinosaurs never evolving occupied a majority of human timelines, then the mediocrity principle would not be satisfied. Thus, in order to keep the mediocrity principle afloat, there must be some kind of detrimental force in the non-dinosaur timelines which is enough to bring it to the minority of total humans. If $\frac{pa}{pa + (1-p)b}$ is the probability of a randomly born human spawning in the timeline where dinosaurs evolve, then the mediocrity principle suggests that $pa > (1-p)b$. Now, momentarily isolating a , the probability of eventual human life in the timeline where dinosaurs evolve, we know that in order for humans to exist in this timeline, the dinosaurs must also then die, because such a thing is a Great Filter. Therefore, it must be so that the dinosaurs die in order for the probability of humans denoted with a to satisfy. Because the Great Filter itself of the death of dinosaurs has a probability of no more than 5%, it follows that $a < 5\%$, otherwise the Great Filter of the dinosaurs' death will no longer be a necessity to human life. Writing this as $a < \frac{1}{20}$, and rearranging the base inequality of comparison of $pa > (1-p)b$ into $\frac{1-p}{p}b < a$ yields a combined result of $\frac{1-p}{p}b < a < \frac{1}{20}$. By the transitive property, this means that it must be so that $\frac{1-p}{p}b < \frac{1}{20}$. (Also note that this means that $20b < \frac{p}{1-p}$). From here, noting how in only one out of three eras of the phanerozoic did anything like a dinosaur evolve, (being the Mesozoic Era), and also since it seemed very unlikely in some of the other eras for this kind of evolution to repeat itself, and thus it may be concluded that $p < \frac{1}{3}$. Then, if this is the case, it follows that $\frac{p}{1-p} < \frac{1}{2}$, and so, again through alignment, $20b < \frac{p}{1-p} < \frac{1}{2}$, meaning that $20b < \frac{1}{2}$. Lastly, it must therefore follow that $b < \frac{1}{40}$ at the very most if the mediocrity principle is to be satisfied. Also, if further factors decreasing the probability of humans in the dinosaur timeline are to be discovered, encompassing some a , then this inequal-

ity may be expanded into the following.

$$b < \frac{a}{2} < \frac{1}{40}$$

Either way, and the result is clear – in order for the mediocrity principle to be satisfied, the likelihood of humans evolving if dinosaurs never came to be is quite low, clearly by means of some happening or side effect of the world in that scenario which severely limits the chances of humanlike beings there. Also note that no implication of b being near to its cap value of $\frac{1}{40}$ has been made here. This inequality shows us that it is equally plausible that b is extremely smaller than this, to the point of the evolution of dinosaurs being a new Prerequisite in its own right. It seems as though this mathematical result is hinting at such a reality, especially considering how through $b < \frac{a}{2} < \frac{1}{40}$ that as we compress upon further discovery the likelihood of the dinosaurs dying through a , a sure thing given its Great Filter status, the devastation factor of the non-dinosaur timeline nose-dives further and further to zero, into the territory a Prerequisite would hint at³. And if it is further determined that perhaps the evolution of dinosaurs was a Fluke, too, and then now it would satisfy the definition of a full Great Filter,

³One key insight to help demonstrate this is the probabilistic nature of the Cenozoic itself. Because the Cenozoic has not yet been multiplicatively factored into these two formulations of the evolution of dinosaurs and death of dinosaurs, it therefore means that it may only be further integrated into the formula as is. Consider it as some additional variable, C for example, which is a factor of a , and is known to only affect a further from what the $\frac{1}{40}$ has already done. Therefore, we may conclude that whatever the standing probability of humans coming to exist at the beginning of the Cenozoic was, the probability of humans coming to exist in an entirely dinosaur-absent timeline is, at most, a striking forty times less than this, in some form of $\frac{C}{40}$. As an example, if we demonstrated that there was less than a one in one hundred chance of the earliest primates of the Cenozoic one day evolving into something humanlike, then we would have to logically conclude, (at least to satisfy the mediocrity principle), that a world without ever having dinosaurs would have less than a one-in-four-thousand chance of generating humans. This is strikingly different then the perhaps intuitive-at-a-glance idea that maybe dodging the probabilistic troubles that are the dinosaurs would have to be very beneficial. The point is, the sky is the limit when it comes to this extra Cenozoic factor, to the point where calling the origin of dinosaurs a Prerequisite becomes not only plausible, but perhaps even probable.

making it a critical but difficult aspect of the road to humanity.

Wrapping up this thought experiment, since we now know that humans would be extremely unlikely to exist if dinosaurs never came into being, the ecological reason for why becomes an apparent question. This is perhaps due to evolutionary pressures from the dinosaurs themselves, which would be absent if the dinosaurs had not come into being during the Triassic period. If this were accurate, then the most feasible dinosaur clade to be the source of this pressure are the theropods, because they were the only ones who likely came into extensive contact with the Mesozoic ancestors of humans. Perhaps there are instances of other dinosaurs affecting human ancestors now and again, but even this would be hardly continuous and direct, to the point where calling this the source of the pressure would be assuming a less likely occurrence, hence subtracting from the human-mass there satisfying. So if theropods are the probable answer to this question, then the next point of interest has to regard what pressure they were applying. Again, there is a most feasible answer, and then a multitude of less consistent ones. That answer regards theropods hunting and killing human ancestors over vast areas and stretches of time, encompassing a maddening display of bloodshed and terror which plagued our ancestors for an unimaginable length of time worldwide. Tracing back the revolutions in our brain during this period, and it can be seen that this seemingly eternal dark period in our history is likely responsible for the sensation of fear itself, along with the steps our brain takes to escape the source of said fear – the mammalian memory, a thorough understanding of one’s placement in time, family bonds, empathy, and the beginnings of human intelligence. Perhaps these vital adaptations in the span of human prehistory were impossible to reach without such an extreme level of horror, and so unveils this new, dark Great Filter. Let the time stretching from this dark Great Filter up until the extinction of the non-avian dinosaurs be known, at least in the scope of these analyses, as *The Dark Era*, to signify its state as the arguably darkest period in human evolution, both in terms of general “quality of life” and literal average perceived illuminance. (Due to the nocturnality of

our ancestors at the time [7]). Note that a hunting theropod is a very different prospect to a small mammal than say a predatory fox or cat, because of superior anatomy and recklessness. If the predatory world was populated by foxes at this time, no cynodont would attain fear because any pressure would be significantly lacking. Only if the absolute most terrifying creatures are continuously seeking out the blood of proto-mammals is there enough pressure to satisfy the Great Filter, and hence dinosaurs had to evolve.

4 The Golden Point

In the history of astrobiology, it has always been useful to categorize alien civilizations under terms like those found in the Kardashev Scale [8], with a Type I civilization representing a society capable of controlling the energy of a planet, Type II controlling a star system, and Type III controlling a galaxy. This may seem intuitive at a glance, because after all they grow in proportion with the different accepted tiers of size in the universe, but it raises something of a question: is there a most fundamental way to categorize civilizations universally? Would such a system be the Kardashev Scale, or something better entirely?

To answer this, consider the civilizations of the universe. Perhaps a better term from here onward is a “biosphere”, because civilization is potentially riddled with bad stereotypes which cloud our vision, and thus are due to be thrown out of science in future iterations, similarly to the geocentric model but even further into the depths of the human experience. That being said, what would an objective observer notice about biospheres universe-wide? Well, firstly, that they are rare, by the Rare Earth Hypothesis. And secondly, that they all must eventually die. Some few live longer than others, but such biospheres only do so out of sheer luck, and have little say in the matter. The organisms making up the biosphere are so concerned with their own individual survival, that they cannot altogether make grand decisions which protect the entire biosphere. If a supernova explodes nearby, and the entire biosphere goes up in flames. That is simply how it works. Let this fundamental

death of a biosphere be referred to as a “Departure Point”, signifying the departure from advancement once and for all. But, statistically, as is expected given a large enough sample size of space, and there should be expected biospheres which learn to protect themselves from Departure Points.

Some will be better than others. In total there must be a full distribution built of some extremely skilled biospheres at fending off Departure Points, and others who are less skilled. Like conservation status for species on Earth, there is no single universal status, but rather a great range. Moreover, the vast majority of biospheres must start with simple origins, through abiogenesis or however these things occur, so it definitely takes some growth for them to ascend to a point of chosen Departure Point dodging, if at all. Therefore, the skill at avoiding Departure Points of a biosphere changes with time, and so the mathematical models should reflect that. This means the best measure has to be the hazard function. Specifically, the skill of a biosphere in preventing its own Departure Point is inversely proportional to its hazard function. Thus, more skilled biospheres will have to have lower instantaneous hazard functions.

Lower hazard function biospheres will be expected to last longer, whereas higher hazard function biospheres will fizzle out much quicker. Biospheres which are more effective in lowering their hazard functions will, on average, last longer. But what of a biosphere which takes this concept to an extreme? What of a biosphere which decreases its hazard exponentially as a very property of its biological nature? Because the chance of surviving t time follows $S(t) = e^{\int_t^0 \lambda(\tau) d\tau}$, where $\lambda(t)$ is the hazard function, or “danger” of the Departure Point for a biosphere at time t , it follows that if $\lambda(t) = a e^{bt}$, where b is a negative number, (to allow for exponential decay of the danger), then $\lim_{t \rightarrow \infty} S(t) = S_{\infty} = e^{\frac{a}{b}}$. Thus, if a biosphere was truly able to decrease its own danger of death exponentially into the far future, and it would have a probability of being completely immortal at $e^{\frac{a}{b}}$. Real biospheres, however, may look a little different from this – for instance they may not be able to keep up this exponential decay in the distant future universe where energy is harder to come by.

But nonetheless, such biospheres will last for truly vast spans of time, and will do so with ease.

These biospheres which are extraordinarily good at longevity shall be called “Golden Biospheres”. They have, for all intents and purposes, beaten death on a biosphere scale. The Departure Point is no longer a factor for them. This then means that they have fundamentally differentiated themselves from the rest of the biospheres of the universe, which still undergo Departure Points like clockwork, so in a way they are as separate from normal biospheres as those biospheres are from lifeless chemistry. Thus, humanity itself would be of the “life” stage in our development, and not the “Golden life” stage, meaning we are more similar categorically to an amoeba than we are to a Golden life form, to which death is no longer a threat.

If a specific large volume of space could be considered, then the number of separate Golden biospheres which come into being in that volume shall grow linearly, because the average rate at which Golden biospheres spawn in it is roughly constant, and the rate at which those biospheres die is immaterial. In contrast, the number of non-Golden biospheres in that same volume of space will grow in a concave-down manner with respect to time. Their rate of formation will also be roughly constant – and in fact probably much higher than the rate at which Golden biospheres spawn because of how much probabilistically easier it is for them to form, but these non-Golden biospheres will also have a rate of death to consider, so during each new time, the net growth in number will be less than that of the previous time, and hence their number will have to be concave-down. Therefore, just by this logic, given enough time, the number of Golden biospheres will infinitely overtake the number of non-Golden biospheres, and they will make up nearly 100% of all biospheres in the cosmos. This brings about a fundamental rule of biospheres beyond a certain advancement: either they will be obsessed with becoming Golden biospheres, or they will die.

Therefore, the fundamental categorization of alien societies is probably less what the Kardashev scale has us believe, and more regarding the fundamental distinction between non-Golden biospheres with Golden biospheres.

One quick and easy approach to actually approximating the number of Golden biospheres relative to non-Golden biospheres in a region of space, is to imagine the rate at which they form, and the rate at which they undergo their Departure Points. For a classification of biospheres based upon a rate of formation / hazard function of formation of L , and a rate of Departure Point happening / hazard function for Departure Point happening of λ , it should be expected that the number of biospheres of such type in a region of space should follow the following relation.

$$N(t + dt) = N(t) \times e^{-\lambda dt} + (1 - e^{-L dt})$$

This is so due to the universe-wide cropping function of preexisting biospheres, (under this particular categorization), by hazard function λ to each, and then also due to the rising of new biospheres from nonorganic matter or whatever phenomenon spawns them, which takes the form of the second additive term. Just the right manipulation yields the result of how many biospheres of a certain type should be expected to be found across the cosmos.

$$N = \frac{L}{\lambda} (1 - e^{-\lambda t})$$

Note that this still assumes a constant hazard function of birth and death for the biospheres in question, at least relative to time. One may substitute their distribution functions of choice into this relation and see a rough approximation of biosphere frequency returned. In the simple concept displayed before, two sets are considered: Golden biospheres, and non-Golden biospheres, which are taken to be a constant category. Clearly, in actuality, things are not quite this simple. But this does not mean it is not an excellent approximation. Golden biospheres defined here with $\lambda = 0$ may not be physically achievable with all probability being considered, however biospheres following $\lambda \approx 0$ very much are possible, and may even be relatively abundant across the infinite cosmos. If a biosphere is truly this close, it seems quite sufficient that they be named a Golden biosphere anyway, for the sake of practicality in the face of our non-perfectly mathematical expecting universe. Perhaps different scholars may disagree in the

coming years, but either way, the general principles throughout this paper are unchanged; simply in one version, the Golden Point is a tangible point to which a biosphere may aim, whereas in another, it is a technically unachievable reference thought of more as a symbol of aim rather than a tangible rank of success. But under the philosophy of naming all noteworthy close biospheres as Golden, (which is subjective, yet non-limiting and therefore wise for it), there should essentially be a cutoff between biospheres failing to meet a certain threshold of λ , and others who succeed, within the scope of the distribution and equation of abundance.

At the next point, one may further declare if they wish, that the set of non-Golden biospheres are to also be subjectively limited just as the Golden biospheres were, because after all, to define a biosphere, one must first define life, or perhaps more generally, Golden-based-success. Perhaps an arbitrary list of terms will be displayed which enable a biosphere to be deemed a “civilization”, or furthermore, “intelligent” and so regardless of what these terms are, they must significantly lower the count of acceptable biospheres from “total” biospheres. Therefore, the non-Golden biosphere set is substantially reduced, whereas the Golden biosphere set remains untouched, and so whatever high proportions of life the Golden biospheres may have made up already is now furthered significantly. The greater the accepting set of biospheres, the less proportion Golden biospheres must make up, so calling, say, amino acids intelligent-biosphere qualification versus calling spaceships intelligent-biosphere qualification is a vast leap of difference, all contributing to the relative abundance of Golden biospheres.

Furthermore, perhaps the quantity of L is indeed not fixed with time, and instead lessens significantly as the resources of our universe run dry with age. In fact, the predictions made later in this paper, which, among other things, argue that Golden biospheres form in roughly similar ways and during roughly similar timeframes, would mean that the falloff is sudden and intense for the cosmic value of L , at least in our general location, as things head for the worse over time, (whatever that is, anyway). Perhaps time is immaterial to the infinite expanses of the cosmos,

in which case, just like over distance, certain high-success areas significantly outpace low-success areas, (the classic example being Earth and Mars), making entropy the sole driver of where life can be found, and with it the mediocrity principle’s mysterious power. The factors of Golden biosphere appearance similarities, significant reduction of biosphere qualifications for the sake of comparison, and the eventual decrease of life generation over time, all make the cosmic proportion of Golden biospheres noteworthy higher even than what that equation of $N = \frac{L}{\lambda} (1 - e^{-\lambda t})$ can explain, and it already has a tendency to favor Golden biospheres over the alternative as is. (Unless an absurdly variant Departure Point probability distribution is substituted). These factors, collectively, are what make the Golden biosphere versus non-Golden biosphere binary such a fair approximation of relative cosmic abundance.

As for what such a categorical transition would look like in practice, a great framework is to imagine the total Departure Point as being composed of two distinct subcategories, those being the “Internal” Departure Point, (caused within a biosphere), or an “External” Departure Point, (inflicted on a biosphere from outside). The tools which a biosphere uses to prevent these respective Departure Points may often differ, so focusing on one of them alone will still, statistically, prove to be disastrous. The key to classifying as a Golden biosphere is to be masterful at lessening the probability of both types of Departure Points simultaneously rather than only one or none at all. The strategies between them may be separate, so enlightening a biosphere in the art of all these strategies in a harmonious whole is the key to upping one’s classification in the scheme of cosmic life. This automatically means that power and domination is likely inefficient because it feeds the chances of the Internal Departure Point. While the exact set of choices to become a Golden biosphere may be as of yet unclear, it seems as though a combination of the essence of technology with more Daoist ideals is in fact closer to true efficiency than any patriarchal system will be in the long term.

The point at which a non-Golden biosphere ascends to a Golden biosphere is called “The Golden Point”, and it can be considered the moment when

the progress of a biosphere is growing too quickly to allow the Departure Point to happen, similarly to how a Hubble horizon marks the point at which spacetime is expanding too quickly for travel across it to be possible. Thus, perhaps a definition for the Golden Point can be, “the point after which a biosphere becomes too capable at self preservation to regress past that point.” There may in fact be several points which satisfy this property, or even a continuous ray of them, making a “Golden ray”. In this scenario, then, the official Golden Point would simply be the first of these points.

Using this logic, the Kardashev Scale now becomes perhaps even faulty, because it threatens to contradict the Golden Point. A Type I Civilization is still more or less accurate – after all they exist in the foreseeable future for humanity at least, and may even have some kind of linkage to Golden biospheres. Once a biosphere can sufficiently control its own homeworld, and that means it will be able to edit its own environment, similarly to how humanity can today, but to a far more spectacular degree. At some point this is bound to initiate a feedback loop, as the biosphere augments the use of its own success in fueling future success. This is when the Golden Point is reached. Beyond this point, however, there is no telling what could happen. However, unlikely occurrences are still easy enough to spot. Take, for instance, a Type II and a Type III Civilization. Why do we assume that a Golden biosphere would seek to control something? That is the work of modern humans, and if anything it hints at our primitive nature with respect to a Golden biosphere, rather than the alternative. The will of modern humans to control, dominate, and draw the resources out of our environments is patriarchal and altogether ignorant with respect to more objective ways of thought. The arrival at such objectivity would be long behind us if we ever were to become a Golden biosphere. And if a Golden biosphere was truly concerned with drawing the energy out of its environment rather than integrating itself with that energy, or some other foreign method of life which we cannot yet comprehend, and it could potentially seek to draw energy out of itself as well, as we can clearly see humans doing in the modern day, (in wars, pollution, and extinction driving,

etcetera). The moral of this story is that our will to divide land amongst ourselves today is biased enough that a Golden biosphere might not necessarily think the same way, and if it did, there is a good argument to be made that such a mindset would contradict the Golden Point’s definition anyway, because it would make room for that biosphere to destroy itself.

Lastly is the prospect of humanity’s future. Many futurists believe we are on our way to becoming a Type I Civilization, and if that thesis is accurate, it means there is a noteworthy chance of us becoming a Golden biosphere, as well. Amara’s law alone tells us this – with the Type I Civilization representing humanity’s prediction for the future, and the Golden Point representing the true reality which we had vastly underestimated with our prediction. But let us take a step back, for a moment, and consider what this would mean. There is often a bias in human thought which thinks of the human era as all there ever was in the timeline. But then one learns of deep time. Deep time is the history of the Earth which stretches backwards in unimaginable leaps, seeming to lose itself in the fog after a while. From our perspective, the Earth is practically eternal. It is incomprehensibly ancient. By the logic distinguishing Golden biospheres from non-Golden biospheres, life on Earth has been on a fierce mission to reach its Golden Point that entire time, beating shocking odds and miraculously avoiding the Departure Point through sheer luck. Hence, the Rare Earth Hypothesis. But now all that has a purpose – to become Golden Life, or die trying. Sure, life may not consciously choose to do this, but in the set of all Golden Life, their predecessors have something of a trend of making the exact right decisions at the exact right times on the road to the Golden Point. And now, again, the question comes to humanity. We are not some simple fluctuation in a short-lived reality – rather we have just been through an astounding triumph over five billion years which has brought us to the very gates of the Golden Point. If the entire trek to the Golden Point could be compressed into a single year, the beginning occupying the birth of life, and the end taking the form of the Golden Point, and we, the humans, at this very moment, would be within the last second before the ball drops on New Year’s

Eve. This type of mindset, if accurate, might as well render the Copernican Principle obsolete, because of how extraordinary and eerie it is regarding our purpose in the universe. There is something very weird indeed going on. Why were we not born in the dead center of deep time, or closer to the beginning, or anywhere more moderate for that matter? The odds of being born within the last closing second of the allegorical year is beyond remarkable, and yet here we are. I have yet to hear anything remotely as remarkable in my life. (I would like to think this is not the fatal flaw of the mediocrity principle, but rather that there is some unknown force of nature which skews things toward the time directly preceding the Golden Point. But whatever the cause, it is an incredible feat of nature no different than the other unsolved questions of mathematics and physics, and thus needs to be treated as such. It seems as though this question and the question of why entropy is so staggeringly low in the modern universe are in fact two sides of the same coin, and will someday be answered together).

Let this newly identified concept be referred to as “The Golden Principle”. The Golden Principle states essentially four main points:

1. All life either tends toward the Golden Point, or is destroyed
2. Humanity is remarkably close to the Golden Point temporally
3. Rather than humanity’s observations being representative of that of the average position in the universe as the Copernican Principle states, humanity’s observations are much closer to being representative of that of the average Golden Point biosphere in the universe
4. There is some unknown form of causal (or causal mimicking) connection between our existence in the time and place we live with the fundamental concept of the Golden Point
 - Let this unknown connection be referred to as “The Trace Connection” until it is officially derived

(As a rough application of this principle to understand how it approaches the meaning of life on Earth, if we somehow discovered that humanity was in a simulation, then the Golden Principle would logically conclude that the simulation is in fact a Golden Point simulation. Obviously we may very well not be in a simulation as well, and if that is the case, then the Golden Principle would argue for whatever general translation from a Golden Point simulation to our actual reality would be. If we are not in a simulation, then whatever it is that’s actually going on can only be more remarkable.)

Lastly, if the Golden Ray is to be all time existing after the Golden Point, then the Golden Trek is all time leading up to it. We, currently, then, are located at the pinpoint end of the Golden Trek, following immortal work done by our ancestors and circumstances leading back through the deep swaths of time. The decisions even of a proto-mammal living in a burrow under the theropod ruled Earth are as vital to the Golden Trek as those of a sentient human living today. So, in a word, humans have not started the story, but rather we must choose to finish it for the sake of the Golden Point.

5 The First Law of Auruspectation

Now it is understood that the mediocrity principle is better applied to the set of all possible humans in all possible alternate realities rather than simply the humans which existed visibly in our own history. But this human-centric approach has its own problems, specifically regarding the arbitrary nature of what makes a human a human in the first place. Where is the biological cutoff point between being born a human versus not? Why does your next door neighbor count as a human, but a chimpanzee does not? What is the arbiter of that distinction?

For this question, the Golden Principle itself may be of use. If the Golden Principle is valid, then humanity’s observational reference on the universe serves as a good approximation for that of the average Golden Point biosphere, so by that logic just

as one can identify themselves as a human, they can also approximately identify themselves as an element of a Golden Point biosphere, at least in the observational sense. Clearly there are major distinctions between humans and Golden Life, but in the scope of the outward observation of the cosmos, one can reliably make this approximation so long as the Golden Principle is valid. If, then, one was an element of a Golden Point biosphere, it would make sense to apply the same mediocrity principle logic as before. Rather than picking a random human body to be born in, now one may mathematically consider a random Golden Point biosphere to be born in, which then corresponds to its preceding Golden Trek.

Essentially, from this, one may consider themselves as being born within a random Golden Trek in the set of all possible Golden Treks in all possible alternate realities. This is the First Law of Auruspectation: consider oneself as an element of a random (successful) Golden Trek in the set of all Golden Treks. As before, the mediocrity principle can now be applied in the same way, making sure not to link it to every single comparison that is considered, but rather to each one individually with a high chance of accuracy. Using this, one will arrive at the same conclusions as before regarding for instance the high chance of being born in a world where dinosaurs evolve, and then go extinct, rather than never evolving or never going extinct.

Mathematically speaking, the probability of an Attribute in the set of all outcomes might be p , however in the set of all Golden Points this then becomes, using the same logic from earlier, the following.

$$g = \frac{pa}{pa + (1-p)b}$$

Because of the Golden Principle and the First Law of Auruspectation, it can now be determined that g is an extremely good approximation of the probability of an unknown Attribute of the universe being in effect as opposed to not. Note that it is still perhaps not perfect, especially in specific conditions which exaggerate the distinction between the modern day and the Golden Point, but it is still worth noting that for any Attribute older than say ~ 1 million years in origin, (which is sufficiently far before the evolution of

humans, itself a Prerequisite which potentially sways the probability of things Golden-ward), this law will hold with striking precision. Note that g is referred to as the “Golden Probability”, whereas p is the “Outlook Probability”.

In short, Outlook Probability is the likelihood of the Attribute in the set of all possibilities, and Golden Probability can be taken as the likelihood in the scope of human observation, as pertaining to the Golden Principle in place of the Copernican Principle.

6 The Second Law of Auruspectation

The Second Law of Auruspectation is perhaps a useful tool regarding what happens when a particular Attribute is statistically independent from the Golden Point.

If some Attribute is statistically independent from the Golden Point, then that means it will have no net effect on it. This means that the Attribute could be flipped to an exact opposite condition, and still the Golden Point’s probability would remain unchanged. In accordance with the mathematical form of the First Law, $g = \frac{pa}{pa + (1-p)b}$, this means that $a = b$, because the Golden Point’s chances remain untouched as a function of the Attribute alone. If some variable x is substituted for them, such that $x = a = b$, this means that $g = \frac{px}{px + (1-p)x}$. Factoring out x and simplifying yields the mathematical form of the Second Law – but only if the prior condition of the statistical independence is met.

$$P(A \cap G) = P(A) \cdot P(G) \rightarrow g = p$$

This will be applicable if A represents the Attribute and G represents the Golden Point. In a written form, this states that *if an Attribute of the Golden Trek is statistically independent to the Golden Point, then its Golden Probability equates to its Outlook Probability.*

As an example of where such a concept becomes useful, let us consider the dinosaurs once more, and specifically ask what effect the asteroid had on their

extinction. Was it the major cause of the devastation, one unextraordinary cause among many factors, or perhaps the sole cause? To get at this answer, it helps to first consider what might be statistically independent to the Golden Point, or, assuming the validity of the Golden Principle, humanity. The asteroid itself wiped species out on the basis of how much metabolic energy they needed, or if they could burrow, resist cold temperatures, hibernate, etcetera [9]. Such things are, first and foremost, dependent on body mass. There is perhaps a limiting function which puts the extinction hazard upon a species as a function of weight, and due to the vast growth of exponentials, will create a sort of cutoff weight at some point or another which practically guarantees extinction. Because of this, no matter how successful the dinosaurs might have been directly prior to their extinction, they would still collectively face the same danger in the event of the asteroid because they would still be above the cutoff weight. And on an even more specific note than weight, they would still lack the biological requirements to even remotely survive that type of event and aftermath that is the K-Pg extinction. Hence, the well-being of the dinosaurs directly prior to the extinction event is nearly statistically independent to the chances of them being wiped out by the asteroid itself. And because humanity is almost fully dependent on that extinction by virtue of it being a Prerequisite, the net linkage means that the wellbeing of the dinosaurs directly before the extinction event is nearly statistically independent to the chances of humanity ever coming to exist.

This is where the Second Law itself becomes useful. By the logic of the law, it follows that the Outlook probability distribution regarding the well-being of the dinosaurs directly prior to the extinction matches its Golden probability distribution. The question, then, becomes which is more probable – the dinosaurs staying at roughly the same level of success as they always had throughout the Late Cretaceous, or taking a mysterious dip in this success as a slow dinosaur death model might suggest. While technically both are possible, it makes much more sense that the dinosaurs would stay at roughly the same level of success up until the moment of the extinction, because anything on the contrary would be uncharacteristic of

them. Truly, the whole point of the Great Filter that was the mass extinction was to wipe them from the Earth, so if they were going to take a dip in success anyway during the Late Cretaceous, when literally anything could happen to them (by the application of the Second Law), then surely their presence on the Earth would not be so extremely overbearing in the first place. This contradicts a whole list of other things, and hence it becomes the most logical conclusion that if dinosaurs were temporarily free of further effect from all Optimisation⁴, as they were at the very closing of the Late Cretaceous, then they would continue to thrive as they were built to do, and hence their eventual extinction hit them when they were at their height, not their low.

Another great use of the Second Law, is in the context of Solar System bodies, and the extent of their similarity to Earth. If one asks whether or not Venus contains the same general rocks and minerals in its bulk as Earth, then the answer would be almost certainly, because they in fact were made from the same material at the birth of the Solar System. If this were not true, and there would be major repercussions in the chances of the Golden Point, so therefore we give this a low Golden Probability. (Which is why we very much don't see it in reality). But another question may be asked which has a very different result: is Venus naturally habitable to humans? Well, it can be easily reasoned that natural human habitability on Venus would be statistically independent to the Golden Point, and furthermore our chances of coming to exist in the universe. Whether Venus could sustain humans or not, and the Earth itself would be unaffected, so therefore our lives and history on this planet would remain constant. Because of this, by the Second Law, it can be reasoned that Venus's Golden Probability of being human habitable matches its Outlook Probability. And attaining that Outlook Probability is easy through the Rare Earth Hypothesis – it tells us the probability is extremely low. Therefore the Golden Probability, (which remember is the probability in the scope of human observation), becomes quite low as well. Therefore you

⁴Optimisation generally meaning the act of skewing Outlook Probability to Golden Probability in this case. See the Optimisation section for a more in-depth description.

will expect Venus to be inhospitable to human life without technological interference.

7 The Third Law of Auruspectation

Given the Rare Earth Hypothesis, two things become clear. Firstly, the most obvious one, that life in the universe is spatially distant from the nearest separate biospheres. But there is perhaps an additional property that would be expected to result from this type of reality – that Golden Treks themselves are distant in probability from one another. Specifically, small changes in the Attributes of Golden Treks map to huge differences in feasibility of chance. This is the “High Contrast Earth Hypothesis”.

If the High Contrast Earth Hypothesis were directly wrong, then Golden Treks would be expected to be mostly clustered around a single location in probability, with stragglers as there always must be, but uncommon stragglers. This seems unintuitive given the vast multiplicative breadth in probability implied by the Rare Earth Hypothesis. The more intuitive result, then, would be for the probabilities to be more spread out, and hence the High Contrast Earth Hypothesis. We already know that a Golden planet is far rarer than a general planet by the Rare Earth Hypothesis, so with the High Contrast Earth Hypothesis, it would be additionally noted that skewing certain elements of that Golden planet would substantially lower its likelihood of reaching the Golden Point at all. For instance, as we have already seen, if dinosaurs never did evolve, and the likelihood of the Golden Point, relative to the Earth in the early paleocene, would drop by over forty times. A single Attribute change which holds massive consequences.

Such consequences shall be called, “Imperfection Magnitude Drops”. Imperfection Magnitude Drops are, by nature, unsuspecting and extreme. Take a red dwarf star for instance. What is the likelihood of a Golden biosphere evolving exclusively around this star? Such a thing is not truly known, but the High Contrast Earth Hypothesis has its own formula for it nonetheless. The Attribute of difference here from

our own Golden Trek is specifically the mass and temperature of the star. While our star is white and one solar mass, a red dwarf is red and no bigger than half a solar mass. This is a key difference, and vitally, it would have branched off from our Golden Trek at a very primitive stage in our history. The High Contrast Earth Hypothesis would then, by its logic, suggest an extremely high likelihood of there being extreme contrast here. More recent divergences would have this property less so, or at least in likelihood be less of a contrast. The actual logic of the HCEH is that rolling the dice of life with a white star yields a vastly small number, so if another dice roll was made that early in the Golden Trek, and yet again it would likely be a vastly small number, however it would probably not be close to the other vastly small number. If a number is close enough to zero, it could be a trillion times less than another number, and even that bigger number would be extremely close to zero. In fact, multiplicatively speaking, the space close to zero is like an infinitely stretching expanse, so if a white star was randomly placed somewhere on this expanse, and a red star was also, they would be extremely likely to be quite multiplicatively distant from one another. According to the HCEH, at least, the more extreme the Rare Earth Hypothesis, the more extreme this multiplicative contrast would be.

One thing the HCEH cannot independently measure is which star is the higher likelihood of the two. The red star could have a higher likelihood of the Golden Point, or the white star could, and through the HCEH alone one could never know which is which – all it tells us is that whichever one is greater, it is vastly greater multiplicatively than the other one.

To actually order the probabilities as lesser and greater, the First Law of Auruspectation itself is needed. The First Law states that humanity and the Earth represent a random sample from the set of all possible Golden Treks. To truly understand this phenomenon comparatively, it would be helpful to have a system of ranking Golden Treks by how efficient they are in their endeavors. Towards the more efficient end of the scale are Golden Treks which consistently take the most optimal paths in the direction of the Golden Point, and on the lower side of the scale

are Golden Treks with a greater likelihood of failure and a less efficient model, like quantum fluctuations leading to a Golden Point. For instance, the best of the Golden Treks which wiped out their dinosaurs are more Optimal than the best of the Golden Treks which let them live. Their probabilities are higher, and they make up a larger portion of total Golden Points. The actual “mass” in the total Golden Point that an individual Golden Trek makes up is its “Optimality”, a technically infinitesimal value, but one holding much power and importance to how Golden Treks function. Lower Optimality Golden Treks are more common than higher Optimality ones, but each one of them makes up less of a portion of Golden Points due to their low efficiency. Every Golden Trek has an Optimality, and also a specific percentile in the scale from least Optimality to greatest, and because the set of all Golden Treks is a constant phenomenon, percentile and Optimality can technically be written as a direct injective function from one to the other, however such a function is obviously not yet known to humanity in this context. A key consideration, now, regards how the average white star Golden Trek must be located somewhere on this percentile scale, and also how the average red star Golden Trek must be someplace else.

It is known from the High Contrast Earth Hypothesis how the ratio between the Optimality of these Golden Treks will be extreme, so no matter how good a percentile the more Optimal one will have, the less Optimal one will still fall in the lower depths of the first percentile. This is where the logic of the First Law comes into play. If the average red dwarf Golden Trek was, say, at the 80th percentile, then the white star Golden Treks would fall to minute numbers, written in thousands or millionths of a percentile. This situation can instantly be labeled incorrect, because it would directly violate the First Law of Auraspection. If we truly were born in a random Golden Trek, then the likelihood of being born at such a vastly small percentile would be next to impossible, and hence red dwarf stars cannot be the greater of the two. On the other hand, if white stars were at a random percentile, like say the 80th percentile themselves, and the state of red stars sitting at the lower side of the first percentile would not stir up

any contradictions with the mediocrity principle and the First Law, because we, in fact, are not in orbit around a red dwarf star.

The same logic can be shown for blue stars, and perhaps with even more certainty this time. Blue stars are so short lived, that such a thing alone makes no time for a Golden Trek to progress at its usual slow pace. This, again, sends them to extremely low Optimality and percentile. Both of these cases are examples of Imperfection Magnitude Drops, and they illustrate nicely just how devastating such an event is to a Golden Trek. This also shows that all the most Optimal Golden Treks, occupying the more efficient end of the spectrum, in fact orbit around white stars, because anything differing enough from that would directly lead to Imperfection Magnitude Drops.

And, according to the HCEH, could-have-been Imperfection Magnitude Drops are widespread throughout our Golden Trek. Anything ranging from the diameter of the Earth, to the metals available in the early Solar System, to even something as minute as the angle at which the asteroid which killed the dinosaurs impacted, and all of them must be just about perfectly right in order to dodge Imperfection Magnitude Drops. This in fact allows for a better definition of Prerequisite: *A Mission which, if failed, would result directly in an Imperfection Magnitude Drop of the Golden Mission.* (See the terms page for definitions of some of the included terms).

As a guide for spotting Imperfection Magnitude Drops, (assuming the validity of the High Contrast Earth Hypothesis), they are typically more common as a function of the amount of ramifications an Attribute change from our own Golden Trek kicks up. For instance, with the red dwarf star example from before, the ramifications are overwhelming in an instant. First of all, in compiling the elements of its Optimality, one must choose between putting it quite close to its home star to allow for liquid water, (because red dwarf stars are so cool), and then suffering the consequences of tidal locking, or moving it further away to dodge the tidal locking, but then having the planet freeze over by lack of warmth. Both a tidally locked planet and a frozen planet are Imperfection Magnitude Drops in their own right, this time to perhaps an even more obvious degree than a red dwarf

star. Simply put, a frozen planet has no liquid water, and thus cannot undergo even the simplest aspects of biology, and tidally locked planets are arguably even worse off – the reason being one side is hot enough to boil anything organic in an instant, and the other side is cold enough to freeze it solid. There is a small band in the direct intersection of these two hellish hemispheres where liquid water is potentially possible, but this thin environment is less like the paradise we know on Earth today, and more resembling an industrial nightmare as the weather patterns from both adjacent hells wash over it continuously from both directions. One might notice that a red dwarf Golden Planet, a tidally locked Golden Planet, and a frozen Golden Planet all have one interesting aspect in common regarding their Optimality: they are a mix between an almost definite Imperfection Magnitude Drop, and then an eye-of-a-needle sized “loop-hole” so to speak which seemingly allows escape, such as the thin band around a tidally locked world, (or better yet a binary planet partner which pulls the Golden Planet out of its tidal locking), or maybe a nearby gas giant near the frozen Golden Planet which allows a Europa-like ocean to form under the surface. These loopholes may seem promising at first, until that is we pursue them, when we get the same general pattern – a broad and dominating Imperfection Magnitude Drop in most of the scenarios, together with an even tinier loophole next to it. The more loopholes one accumulates in their Attribute set, the tinier the chances of the Golden Point becomes in an exponential fashion, and the harder it is to find sufficient loopholes thereon. Eventually, using the limited power of the human brain alone, one would hit a wall. But if, hypothetically, the simulation runner had sufficient omniscience to run this indefinitely, until the very most Optimal red dwarf Golden Trek was found, and they would discover essentially a perpetually downward spiraling weighted average of Imperfection Magnitude Drops, with the very best uncovered paths being nothing more than glorified Imperfection Magnitude Drops themselves on account of their poor likelihoods and overly complicated inefficiency. The Attributes of a Golden Trek are a very fluid thing, so trying to find remotely efficient red dwarf Golden Treks is analogous to trying

to carve a word into the surface of a lake – it is far too fluid and ever changing to allow you to.

This same pattern of out-of-control ramifications is easily visible in many other Prerequisite-failing scenarios. For instance, with a Solar System absent of metal, now the raw materials needed to generate the Golden Trek are absent, so one needs to once again descend down a rabbit hole in order to pick up the best salvageable Optimality. The rabbit hole is typically a sign of an Imperfection Magnitude Drop. At least, by the High Contrast Earth Hypothesis, if the rabbit hole tends in a sufficiently distinct direction from our own Golden Trek, then it is expected to achieve a very different dice roll in Optimality, and hence the argument for that.

As for occurrences which are probably not Imperfection Magnitude Drops, it is the art of trying to do the opposite. Rather than coming up with Golden Treks which tend to diverge from our own, this time it is about making differences which still converge roughly in parallel with our own Golden Trek. For instance, if it rained an extra day during the Mesozoic Era, or if there was one less island in the Pacific ocean. The wiggle room here is bearable this time, as there are almost certainly sub-scenarios within these broader Golden Trek sets which have comparable Optimality to the Earth. And hence, in the set of the whole of the Golden Point, these best scenarios will dominate the proportions, making them clear and obvious.

The key conclusions from the High Contrast Earth Hypothesis and the avoidance of Imperfection Magnitude Drops is that our Golden Trek is like an intertwined web of events going perfectly right, and many if not most of the key Attributes therein have the power to instill Imperfection Magnitude Drops in the event of their failure. Golden Treks beyond even the first percentile have to be absent of Imperfection Magnitude Drops to exist, which calls for a strikingly specific set of Attributes which make them up. Think of how difficult it is to name one single non-Imperfection Magnitude Drop Attribute, let alone an infinite array of them. And as the percentile increases beyond even there, the set of all Golden Treks which apply shrinks even more substantially, just limiting the applicable range further and further with height-

ening percentiles.

Let us transition momentarily to the statistical independence or lack thereof scenarios. If an Attribute is statistically independent to the Golden Point, then the results seen through the Second Law documentation will come to pass. But what about Attributes which are the opposite – and instead are very causally connected to the Golden Point, through a kind of statistical dependence rather than independence? Let the quantity of correlation to the Golden Point be called “Golden Linkage”. It will rate on a scale from -1 through 1 , with 0 representing complete independence, with a Golden Linkage with absolute value 1 representing complete correlation, positive being in the positive sense, and negative in the negative sense. Also, correlation will be viewed in a kind of one directional sense, such that it asks for the dependence of the Golden Point on the Attribute, and not of the Attribute through the Golden Point. This way, even if an Attribute holds a low outlook probability of the Golden Point, it will nonetheless have a high Golden Linkage if the Golden Point is sufficiently dependent on it.

As for the mathematical definition of Golden Linkage, it is as follows:

$$L = \frac{a - b}{a + b}$$

This is produced where a is the Outlook Probability of the Golden Point in the presence of some Attribute, and b is the Outlook Probability of the Golden Point in the absence of said Attribute, relative to a specific reference.

And note that it can also be written in terms of the Outlook Probability p and Golden Probability g of the Attribute.

$$L = \frac{g - p}{g - 2gp + p}$$

This all being settled, now let us imagine some particular Attribute of some particular hypothetical Golden Trek. Specifically, the Golden Trek itself is defined as the set of all Golden Treks for which this Attribute is valid within itself, but is absent from the Earth, or more specifically, some separate Attribute which is strictly mutually exclusive to this Attribute

is in fact present in our Golden Trek of Earth. Letting things be defined this way, one natural question is to ask, firstly, just how different such an Attribute of difference really is from our present Earthen reality, and secondly, how great of a contrast this Attribute would create. One simple answer to this question might be that the Attribute itself is hardly very different at all, itself holding a Golden Linkage so very near to zero, it might well have never existed in the first place on Earth, and all would be relatively well. Take, for instance, a rainy day. One might imagine a hypothetical “imaginary” Golden Trek for which all things leading up to this rainy day are identical to that of Earth, but then the simple divergence was made at its event. Where it was rainy for us, what might a Golden Trek where it was sunny instead on that day look like? Note that even defining our Golden Trek in the first place requires finite Attributes, which stubbornly disallows us to perfectly describe it with utmost precision, but perhaps we can take some of these finite Attributes, all most certainly confirmed to be present in the Golden Trek of Earth, and to their set apply the simple event of a sunny day in place of a rainy one, on that particular day with those particular conditions. With this example, the Attribute of difference itself would be the sunny day, whereas its mutually exclusive Earthen reference would be the rainy day which we experienced in its place. For the sake of the analogy, let us say that this Attribute of a rainy day for us was defined to have a Golden Linkage most extremely close to zero. Even a single percent nudge in the positive direction would imply that we are somewhat dependent on it, so let us assume that this dependence is at a practical minimum to begin with. If this is the case, then we might hypothesize that the Outlook Probability of the rainy day was high, because of the logic of the Second Law and because the mediocrity principle would expect us to do so in such a case. However, this is not a requirement, nor should it be. Let such Attributes be named “Outwardly Insignificant”.

Now, to contrast this, there is the consideration that the Attribute actually had high-magnitude Golden Linkage instead, either in the positive or negative direction. Let us say we are unsure about the specific Golden Linkage of the Attribute, but we have

reason to believe it is not low-magnitude like an outwardly insignificant Attribute. That is, perhaps the Attribute operates on a wider scale, or it connects to more things than one simple precipitation or lack thereof. By the High Contrast Earth Hypothesis, the analysis is that as soon as an Attribute rises to the importance of being a part of a web of sorts which connects other potentially important Attributes, themselves being connected to further Attributes, the assessment should be that the importance of that original Attribute is vast because of this exponentiating web, such that taking it down would shatter everything. Take, for instance, the planet Neptune. It may seem immaterial at first, but the fact is that it has some kind of linkage to the Kuiper Belt, which itself has some linkage to the water of the Solar System, the risks of bombarding meteors to Earth, and the boundary into interstellar space. So, just by Neptune's interconnectedness to separate Attributes, it would be wise to conclude that it is not outwardly insignificant.

Using the same logic as before from here, one may conclude that the Golden Probability of the no-Neptune Attribute would be minute, and hence, due to the mediocrity principle, Neptune should have positive Golden Linkage rather than the alternative. This means that Neptune's Golden Probability is likely greater than is no-Neptune's, and thus Neptune's Golden Probability is greater than one half. If P is the probability of this fundamental truth being accurate, then $P > \frac{1}{2}$, assuming the First Law. This is because it is more likely for Neptune to occupy the majority of Golden Probability rather than the minority, because we have indeed observed it.

From here, now let us imagine multiple Attributes in place of just one, and to generalize our P , we shall now identify this probability as the probability of all these Attributes having Golden Probabilities greater than one half, (or in other words, occupying the majority of their respective Golden Probabilities). If one single Attribute has a minority Golden Probability, then the net statement is deemed false. What if the probability of this statement being true, written as P ? Well, clearly, that depends on the Attributes. Let us imagine that all Attributes in question have been observed and proven valid within

Earth's Golden Trek. If these Attributes are not outwardly insignificant, then they in all likelihood have some form of high importance. We may consider, through the High Contrast Earth Hypothesis, that each of these Attributes has some small window of Golden Probability which represents one option, and a vast complementary portion whose ratio to it is huge by the High Contrast Earth Hypothesis. Then, with our very existence, we are born into a random place on each of these Attribute distributions, with a high likelihood of being born in the large portion, and a miniscule probability of being born in the small portion. Thus, over several Attributes, the probability of not being born in a single miniscule section across all of them is taken as the product of all of these large portion Golden Probabilities, which itself remains large for a sufficiently contrasted Earth. As the High Contrast Earth Hypothesis tends to infinity, this effect may be striking. Clearly, though, the contrast of things is still finite rather than infinite, however if it is sufficiently large, it will still feel as some form of infinite. If the Golden Probability of the large portion on these Attributes is a billion times greater than the miniature portion, as it very well might be for many cases when regarding the road to things like intelligent life, then literally millions of such Attributes could all be considered together, and the likelihood of a single one of them placing you or I in its minority portion is statistically impossible. By the HCEH, the contrast explodes in range as one strays further and further from what we see on Earth, meaning there are vastly more examples of each high contrast than each lower contrast. Thus, if one million Attributes were truly to be considered, some few of them may⁵ have noteworthy low contrast, but such Attributes will be vastly outnumbered by Attributes from the insane stretches of supreme contrast, which themselves will be vastly outnumbered by examples of infinitely perpetuating further contrast still.

The point is, while a set of one million Attributes all containing a billion-to-one contrast ratio each may feel unbelievable, this may very easily be the aver-

⁵Even this may be far-fetched by the High Contrast Earth Hypothesis.

age⁶ of things in the very least, especially if the relative ignorance of humanity is the one picking the Attributes. The moral of the story is that even though the mediocrity principle alone is not sufficient to suggest that every Attribute in our reality is at a Golden Probability majority, the High Contrast Earth Hypothesis most certainly is. It suggests, first of all, that Attributes with noteworthily high Golden Linkage magnitude should be extremely likely to have a Golden Probability greater than any disjoint Attributes put up against them, but also that the falloff between attributes like these with outwardly insignificant Attributes is extreme as well. Because the web structure of an Attribute can happen so quickly and so thoroughly as one treads in their thought experiments away from the things we know to have happened here on Earth, and because this web structure speedily threatens Imperfection Magnitude Drops, it makes sense that, in practice, this falloff would happen very suddenly in the spectrum of Attribute difference from Earth. That is, we can somewhat-easily define an outwardly insignificant Attribute, and we can somewhat-easily recognize a very Golden Linked Attribute, but the middle ground is not so easy to come by, to the point of it being perhaps improbable to ever find an Attribute there, so long as the High Contrast Earth Hypothesis holds. If you believe you have identified an Attribute which is there, that is typically a sign that the Attribute is outwardly insignificant. And sure, perhaps the Attribute is outwardly insignificant and had a higher magnitude Golden Linkage than near-zero, for instance a close-call asteroid which swung by the Earth when it may have caused devastation, (this event having low Golden Probability in the process), but these Attributes still do not connect to a myriad of others, and thus do not hold the fate of our Golden Trek on their backs as a more Prerequisite-like event would. Hence, it may be defined as outwardly insignificant⁷.

⁶Geometric mean of course.

⁷Weirder still, outwardly insignificant Attributes like this which substantially threaten the Golden Trek should themselves be extremely difficult to observe both past and present. The mathematical reasons for this will be explained in further detail in the Nescecoins section, but the essential idea is that random threats to the Golden Trek are worrisome enough that

But if all approaches to demonstrate an Attribute as outwardly insignificant have failed, and moreover, the results seem to be pointing at it being more Golden Linked instead, then it is nearly certain that such an Attribute has a Golden Probability greater than any disjoint Attributes put up against it, and for more binary-like states, (rather than distribution-tending), it may be asserted that $g > \frac{1}{2}$ where g is the Golden Probability of the Attribute. Of course, there is also the display to write this in terms of the Outlook Probability, p , the probability of the Golden Point if it were to succeed, a , and the probability of the Golden Point in its failure, b . This, itself, is the Third Law of Auruspectation, due to hold with great precision.

$$\frac{b}{a} < \frac{p}{1-p}$$

Also note that if the Attribute's probability of occurrence, p , is sufficiently low, such that $p \approx 0$, then through Maclaurin Series manipulation it can be approximated that $\frac{b}{a} < p$. This will hold very closely for any Attribute which is a Fluke. Also note that the general formula of $\frac{b}{a} < \frac{p}{1-p}$ may be expressed in terms of Golden Linkage as well.

$$L > 1 - 2p$$

Let this formulation of $L_0 = 1 - 2p$ be defined as the "L-Threshold".

This concept itself is the Third Law of Auruspectation: *Any Earth Attribute which is not outwardly insignificant is most incredibly likely to have a Golden Linkage above the L-Threshold.*

One final realization can be gained from this particular principle, which is that Earth's Golden Trek

the First Law itself wipes them from our observation field. In practice, this means that, historically, Earth has been in about as little danger of imminent destruction as it could have been, all things considered. Lucky avoidance altogether of adversity is more likely to have happened, (with respect to us), than lucky success over materialized adversity. Whatever instances there are of this trend being broken are far too rare for us to ever hope to detect them anyway. This means that there is a general trend for Attributes with 1) scarily high-magnitude Golden Linkage and 2) low connectivity to other Attributes, to have low Golden Probability. Thus, outwardly insignificant Attributes are usually just that – outwardly insignificant.

itself is very close to being perfectly Optimal. Changing any eye-catching Attribute about the Earth, be it its radius, mass, orbit, composition, history, or anything else of the sort, would lead to a weakened Golden Probability. If this property were absolute, then that would by definition make the Earth the single highest Golden Probability series of events possible. Because it is not absolute, however, this will not be perfect, but the principle is near enough to absolute that the Earth is almost exactly Optimal, all things considered. From the set of all Golden Points in the infinite cosmos, if they could be viewed, they would all observationally resemble the Earth to a striking degree, with the exception of rare outliers from time to expansive time.

All of this being more broadly considered, now, a key issue with our modern approach to astrobiology starts to become apparent. In a word, we think of these processes as too discrete and not continuous enough. Many of our equations reflect this, as we seek to add and multiply quantities to get at the broader idea of life in the cosmos. However, such methods tend to rely on a greater degree of accessible information regarding a topic, such as physics or even just practical counting. So, it may be argued, in the scope of our vast ignorance regarding astrobiology, this approach itself is fundamentally flawed. For instance, when we regard the topic of the Red Sky Paradox, we are often prone to ask whether or not the ten-to-one higher abundance of red dwarf stars to white stars like our sun is a defining quality of astrobiology, thus yielding a paradox in the mix when we discover we are not in fact present in this higher-abundance system. And, considering the temporal component as well, (and how long red stars live), it may be argued that the paradox has increased in magnitude even further. However, now, in the context of the High Contrast Earth Hypothesis, this can be determined to no longer be a paradox at all. The main counter argument surrounds the notion that in the scope of something like the HCEH, a mere ten, or even a thousand to one ratio is practically nothingness. I can name several other components that we define as “helpful” to a star system that our sun has but the red stars do not, such as gas giants and a far more stable habitable zone. Maybe this solves the

paradox, then? Well, no. Relative to the HCEH, we are right back where we started. Now we’ve taken a minute ratio, and pushed it back the other way by another minute ratio, while the grand underlying concepts of astrobiology remained strikingly unaffected. The point of the matter is, our discrete meddling is meaningless in the scope of far more extreme ratios of things, which from our perspective would be practically infinite. One way of grasping this is to consider how when we arbitrarily picked our set of properties of life to consider, those being red stars, gas giants, or what have you, we have picked properties which truly are just that — arbitrary. Why is the color or temperature of the star taken as the major factor in the first place? Why not some other attribute of it, or of the system in general? Essentially, here we run into the same issue as what we had with the cosmological mediocrity principle regarding planets — our points of perceived definition are fundamentally arbitrary⁸, and so our results must, logically, reflect that. Therefore, the fundamental best way of analyzing astrobiology should probably not involve simple multiplication and addition of quantities, but instead should encompass the art of relating different systems through their relation to the Earth’s Golden Trek itself, specifically through these three Auruspectation laws already laid out. No independent discrete sets and no assumptions are formulated, and so, at least to the best of our ability, all the most pesky of the progress halting subjectivity of this field has perhaps been diluted. Rather than running the trouble of one’s results being precisely as biased as their chosen inputs, now we can reconfigure the process as a whole to emphasize the points where we have less bias rather than the points where have more, functionally minimizing that bias overall. And so, perhaps the logic of these three Auruspectation laws best extends our pursuit of astrobiological knowledge at this point in time.

⁸...At least on the scale of / relative to the Rare Earth Hypothesis, High Contrast Earth Hypothesis, and cosmic biological happening in general.

8 Doomsday Pacing

One quick and easy application of some of these principles are as pertains to the pacing of our Golden Trek.

There are essentially two types of Departure Points: an internal, and an external. An internal Departure Point is one which is brought about by the actions or lack thereof of factors within a Golden Trek's homeworld itself. For instance, nuclear war or bioweapons in all their unleashed horror could be potential internal Departure Points. As for external Departure Points, they are something of a constant threat from outside the center of the Golden Trek as opposed to inside. The main reason they should be categorized individually is because these external Departure Points are roughly constant in magnitude of threat, because after all the universe is roughly constant in makeup. A planet-killing asteroid is an example of an external Departure Point, as is a nearby supernova or even something more exotic like a strangelet [10], if they even exist. Perhaps the most common of these external Departure Points is simple orbital chaos, triggered by the behemoths of gravitation that are passing stars. The fact that Earth has not been uprooted from the goldilocks zone by this type of disaster thus far is nothing short of Miraculous [11], and hence the external Departure Point threat is increased by it. Now, there are always variations here and there, but it is nonetheless safe to say that over large timescales, the instantaneous threat of the external Departure Point is roughly constant due to roughly constant cosmic phenomena.

This means that given some length of time t , and an external Departure Point hazard function value of λ , the likelihood of the Earth surviving is $e^{-\lambda t}$.

From here, one might consider the prospect of squeezing the Golden Trek to a shorter time frame, altogether shaving off time t from its end. If our Golden Trek wrapped up and triggered the Golden Point t time earlier than it did in actuality, then, due to the nature of hazard functions, we would be precisely $e^{\lambda t}$ times more likely to not undergo an external Departure Point. Because our future will someday have to see either a Golden Point or a Departure Point, it follows that $G + D = 1$, where G and D are the total

likelihoods of the Golden Point and Departure Point respectively. Also, because the Departure Point can be further subdivided precisely into the internal Departure Probability, I , and external Departure Probability, E , the fundamental relation must be so.

$$G + I + E = 1$$

Defining the Earth's values for these quantities as $G_0 + I_0 + E_0 = 1$, their being measured with respect to some arbitrary very distant past reference point, and also defining $G_1 + I_1 + E_1 = 1$ as the values for the shortened Golden Trek allows for manipulation to be possible. To start, because the likelihood of no external Departure Point taking place increases by a factor of $e^{\lambda t}$ due to hazard function logic, it also follows that $\frac{1-E_1}{1-E_0} = e^{\lambda t}$, or likewise $G_1 + I_1 = e^{\lambda t} (G_0 + I_0)$. Solving for G_1 in this case yields $G_1 = e^{\lambda t} G_0 + e^{\lambda t} I_0 - I_1$. By the Third Law, $\frac{b}{a} < \frac{p}{1-p}$, and also in this case, the Outlook Probability p of our Golden Trek occurring as opposed to the shortened one is $p = \frac{a}{a+b}$, because it simply represents the likelihood of our Golden Point occurring rather than the alternative Golden Point, a concept which itself is equal to the likelihood of ours out of the total likelihood of either. Together, this means that it must be the case for $b < a$, and so $G_1 < G_0$. Substitution yields $e^{\lambda t} G_0 + e^{\lambda t} I_0 - I_1 < G_0$, and then further algebraic manipulation returns the result.

$$\frac{I_1}{I_0} > e^{\lambda t} \left(1 + \frac{G_0}{I_0} (1 - e^{\lambda t}) \right)$$

Because t must be a positive number to represent a "positive" amount of time in the past, the quantity of $\frac{G_0}{I_0} (1 - e^{\lambda t})$ must also be a positive number, meaning $1 + \frac{G_0}{I_0} (1 - e^{\lambda t}) > 1$ and thus $\frac{I_1}{I_0} > e^{\lambda t}$ consistently. Therefore, according to this logic, the more time t for which the Golden Trek is shortened, the higher the internal Departure Point probability will grow, and in an exponential fashion at that. For reference, the logarithmic derivative of the lowest possible bound of $\frac{I_1}{I_0}$ is as follows, but in actuality the real-life answer is probably somewhere above that.

$$\frac{\dot{I}_1}{I_1} = \frac{\lambda (G_0 + I_0)}{(G_0 + I_0) - e^{-\lambda t} G_0}$$

This is yet another example of where our Golden Trek is highly Optimal. If the Golden Trek were longer, the external Departure Point likelihood would increase exponentially⁹, and if the Golden Trek were shorter, the internal Departure Point likelihood would increase exponentially. In effect, this situates the actual temporal locations of things in our past to align in the exact perfect spot to fight off these two combating forces. If the background hazard function of the external Departure Point were lower, this most optimal balancing point would move forward in time, because waiting slightly longer to perfect internal probabilities would be met with a less harsh external Departure Point likelihood in the process, since after all it would be slower to increase. Also the falloff of nudging the Golden Point in either temporal direction would be more gradual – or at least the lower bound of what is allowed would be. The rest of all this is essentially a kind of pacing process of situating events in the Golden Trek. The higher the background hazard function of the external Departure Point, the more pacing is required, and the more events must hasten to avoid wasting time. If there were zero threat of the external Departure Point, then hypothetically the events of our Golden Trek would have no time constraint at all, and could take endless time to complete tasks. But because there is an actual threat of the external Departure Point, the events of our past and Golden Trek must quicken accordingly – but not too far, might they have too much cost with respect to benefit, hence feeding the out-of-control internal Departure Point danger existing too far in that direction. What is truly feasible, then, is the middle ground, where both factors are relatively at their lowest potential – and this is just where one might find the Earth.

⁹Note that unlike the internal Departure Point, this does not increase perfectly exponentially, but it will still appear to be exponential under short time-shifts. (Over long periods of time it will level off to one). The wording is for the sake of noticing symmetry.

9 Nescecoins

A useful unit of measurement in the scope of comparing Golden Treks is what I like to call “Nescecoins”. It is difficult if not impossible to imagine the total probabilities involved with Golden Treks, but relative to one another, the ratios of things become tangible. This is where Nescecoins can be used.

The basic idea of a Nescecoin is as simple as it is terrifying: imagine a hypothetical coin which was perfectly Linked to our Golden Trek, (giving it a Golden Linkage of absolute value one). This coin can be flipped, and if it gets tails, the world ends in a blinding flash. If it gets heads, however, nothing negative happens. Depending on the grade of Nescecoin, something positive may or may not happen as a result of getting heads, but the default version simply triggers nothing in particular if it lands on heads. And if there are multiple Nescecoins to be flipped, n of them, say, then the likelihood of the Golden Trek surviving all of those flips is precisely 2^{-n} .

But the standard use of Nescecoins is as a comparison. For instance, because a scenario where dinosaurs never evolved has forty times less of a Golden Point chance at the very most than if they did indeed evolve, it can then be reasoned that the no-dinosaur scenario is probabilistically equivalent to our scenario plus at minimum about 5.3 Nescecoins, (because $5.3 \approx -\log_2(\frac{1}{40})$). Five Nescecoins in and of itself is a scary prospect, because all five must land on heads in order for the Departure Point to perpetually be avoided, and yet this is just the bare minimum of requirements for the non-dinosaur scenario.

One other application of interest for Nescecoins is in imagining a hypothetical Golden Trek which consisted entirely of Nescecoins. Clearly such a thing is impractical in real life, but it nonetheless is perhaps an interesting mathematically-perfect tool for gauging things in our own Golden Trek, similarly to how calling the Earth a sphere is mathematically useful. The rule of this imaginary Golden Trek is simple – similarly to how there is a constant threat of the external Departure Point, the imaginary Golden Trek has a constant hazard function of picking up Nescecoins, which is some value λ . This means that the probability of picking up at least one Nescecoin by

time t beyond the start of the Golden Trek is $1 - e^{-\lambda t}$. So if this is the probability of picking up at least one Nescecoin, then the probability of picking up at least two is $(1 - e^{-\lambda t})^2$, and the probability of picking up at least n is $(1 - e^{-\lambda t})^n$. The probability of picking up at least $n + 1$ is also $(1 - e^{-\lambda t})^{n+1}$, meaning that the probability of picking up exactly n , with no more or no less, is $(1 - e^{-\lambda t})^n - (1 - e^{-\lambda t})^{n+1}$.

Then, because n Nescecoins translate to a probability of the Golden Point of 2^{-n} , the total probability of picking up exactly n Nescecoins and in the end getting a Golden Point out of it is $\left((1 - e^{-\lambda t})^n - (1 - e^{-\lambda t})^{n+1}\right) 2^{-n}$. This then means that the total probability of the Golden Point, taking into account all possible Nescecoin configurations in the process, is $\sum_{k=0}^{\infty} \left((1 - e^{-\lambda t})^k - (1 - e^{-\lambda t})^{k+1}\right) 2^{-k}$. This being said, the probability of being at a randomly chosen Golden Point, (by the First Law), with exactly n Nescecoins having been picked up, (or in other words the Golden Probability of picking up n Nescecoins exactly by the time of the Golden Point), is as follows.

$$\frac{\left((1 - e^{-\lambda t})^n - (1 - e^{-\lambda t})^{n+1}\right) 2^{-n}}{\sum_{k=0}^{\infty} \left((1 - e^{-\lambda t})^k - (1 - e^{-\lambda t})^{k+1}\right) 2^{-k}}$$

The final step here regards the length of the Golden Trek. The observable universe has only a finite age, but because that age is so inconceivably long, it is practically infinite from our perspective, and hence it is more accurate than not to take the limit of the Golden Probability of having n Nescecoins as t tends to infinity.

$$g(n) = \lim_{t \rightarrow \infty} \frac{\left((1 - e^{-\lambda t})^n - (1 - e^{-\lambda t})^{n+1}\right) 2^{-n}}{\sum_{k=0}^{\infty} \left((1 - e^{-\lambda t})^k - (1 - e^{-\lambda t})^{k+1}\right) 2^{-k}}$$

This is the Golden Probability of n Nescecoins exactly. Interestingly enough, it simplifies to a much simpler form.

$$g(n) = 2^{-(n+1)}$$

This means that, amazingly, in such a Golden Trek, half of all Golden Points have zero Nescecoins picked up whatsoever, and then an additional fourth have only one Nescecoin, and then an eighth have two, a sixteenth have three, and so on. Perhaps this is analogous to our own Golden Trek in a way, because an Imperfection Magnitude Drop, which relative to zero Nescecoins is, say, above twenty Nescecoins in count, and yet in this model such Golden Points only make up about 480 billionths of all the possible Golden Treks, hence showing how infeasible Imperfection Magnitude Drops are, and especially how quickly such things can descend into those kinds of states¹⁰.

10 Importance of Hațeg Island

One surprisingly critical element of the Mesozoic, at least according to the logic of the Third Law, is Hațeg Island [12], an island landmass free of dangerous theropod dinosaurs toward the end of the Cretaceous¹¹. What makes this location so noteworthy at a glance is that it not only represents a safe-haven from the menacing dangers of the Dark Era, but it also seems to be, if not a Fluke itself, on the more unlikely side of things as opposed to the more likely side. Other dinosaurs were present on the island, so why were threatening¹² theropods not? Moreover, there

¹⁰This seems to be evidence for the High Contrast Earth Hypothesis, as well.

¹¹It is certainly worth noting that the exact carnivorous ecological makeup of Hațeg Island is still in some ways speculative, however the abundance of miniature dinosaurs [12] with respect to their mainland equivalents is if nothing else a sign of decreased predatory danger overall. (And of course the presence of *Zalmoxes* [12] can only further this argument, because of the clear lack of fitness of iguanodonts to survive on the mainland at the time – as in they were literally extinct on the mainland [13]). The arguments of this paper are logical offshoots of the notion that theropods of any semblance of hazard with respect to what was going on on the mainland were utterly absent from Hațeg Island's ecological makeup. This could be due to an inherent inability to reach the island, probably combined with a significant amount of, to put it bluntly, luck.

¹²Threatening here being defined as roughly analogous to species which are flightless and or have no flying ancestral species.

was a mammal population on the island, which automatically makes those animals just about the safest in the Mesozoic world by a wide margin. Something like *Hatzegopteryx* or even smaller pterosaurs are not remotely as menacing as the most threatening of mammal-hunting theropods, because of their comparative lack of anatomical nightmarishness. Thus mammals living here would be under a temporary escape from the horrors of the Dark Era, allowing them to undergo their own forms of the island effect and altogether live vastly more peaceful lives. This, however, potentially threatens them even further in a way, because if any hostile theropods were able to make it to the island, through rafting or some other mechanism, then the mammals there would be ill-equipped to survive the rampage which would follow, similarly to the dodo bird. In effect this yields a whole multitude of reasons why the continued survival of Hațeg Island mammals was unlikely, and why the perpetual lack of theropods there was so anomalous indeed. (Especially considering how it happened directly at the end of the Mesozoic rather than some comparatively arbitrary time).

In fact, using these properties, and the Third Law of Auruspectation, a very interesting conclusion can be made about Hațeg Island in general. Considering how it is about the same size as Hispaniola [12], if Hispaniola were the only place in the world holding some specific property, then because its area is about one percent the area of all the world's non-continental islands, this means that if it were truly the only Mesozoic case (for an island its general size) of a fully theropod-lacking island, then there would, statistically, be a ballpark no more than a 1% chance of such a situation ever coming to pass in terms of Outlook Probability. Thus, $p < 1\%$. Feeding this into the L-Threshold from here yields a result of $L_0 = 1 - 2 \times 1\%$, which simplifies to $L_0 = 98\%$. From here, the Linkage to the Golden Point must be greater than this threshold, at $L > 98\%$. This is a most intense Linkage to the Golden Point¹³, and far from statistically

independent¹⁴. Hence, just from this logic, according to Auruspectation, Hațeg Island is actually crucial to the chance of the Golden Point: if not a Prerequisite, it is at least a near-Prerequisite. In other words, without Hațeg Island, you and I would likely not be here.

So why is this island so important? One lead in answering this involves the evolution of Boreoeutheria through the first Primates [14]. This lineage roughly coincides with the timeframes of Hațeg Island, as well as the proposed hemisphere of things, so perhaps there is something of a linkage here between the two. This of course means that human-ancestors of the time would be present on Hațeg Island at the closing of the Mesozoic, during at the very least the Maastrichtian, asteroid strike, and opening of the Danian. Without Hațeg Island, there would be something of a contradiction between the horrendous conditions of the Dark Era, and the delicate arboreal lifestyle expected of proto-primates, even though the two should slightly overlap. Hațeg Island, now, can very well explain this overlap by stating that the earliest primates were in fact in a safe haven from the theropods. Maybe primates having a head-start on other arboreal animals at the beginning of the Cenozoic was key to their later success at controlling the niche, and

value itself is quite a bit lower than the 98%, but it would be statistically wise to imagine a slight upward buffer beyond that 88% lower bound. But, even without such an additional boost toward high Golden Linkage, 88% Golden Linkage is still certainly some sort of eye-catching by itself, because of its viscosity to 100% Golden Linkage, after all.

¹⁴One might argue that in order to run the Third Law in the first place, one condition is that the Attribute must not be statistically independent from the Golden Point, so the conformation of that itself is circular logic. However, the rebuttal to this is simple: Third Law aside, the likelihood of the Attribute being statistically independent in the first place is slim, because if it were truly independent, by the Second Law its Golden Probability would be about 1%, and hence it would be ridiculously unlikely for us to see it in the fossil record at all. From there, by the High Contrast Earth Hypothesis, there is an immediate and harsh falloff from statistically independent to very much Linked, hence allowing the application of the Third Law, (in the likely condition that the Attribute is not independent), which concludes that the Attribute is very Linked at the bare minimum. Not to mention, on the scale of Hațeg, during such a defining period of Earth's history at that, it is quite additionally infeasible by the HCEH of the statistical independence at all. Thus, this logic is justified.

¹³Note that this area method is far from precise, as approximations go. All this being said, towards the end of the Temporal Third Law section, a more reliable lower bound is set, at a Golden Linkage of 88%, using the logic of the likelihood of avoiding oceanic dispersal of hazardous theropods. This

perhaps doing so in the northern hemisphere, (where it had been spring [15] during the asteroid strike), further helped their longevity. All things considered, without Hațeg Island, perhaps many of the primate attributes we take for granted would not have come to pass, or instead would have gone to some lesser clade which would have extreme comparative probabilistic difficulty evolving into humans. Thus, Hațeg Island and primates are potentially very closely connected, at least according to what Aurspectative Reasoning has to say. This is a kind of “Primates as Mesozoic Dodo Birds” hypothesis.

And the last thing to consider here is the event of the K-Pg extinction itself, and how this applied to our lineage. Logically, some species must instate a Golden Point, and logically, this species must possess a direct ancestral species which survived the K-Pg extinction event. Therefore, such an ancestral species must be very fit at surviving such an extinction, especially relative to certain other species, such as the set of non-avian theropods themselves. We can imagine, arguendo, a hypothetical scenario where every species was equally fit at surviving the extinction, but then it would not make sense why so many separate lineages of, say, placental mammals survived, why other entire clades did not. Not only would this be statistically improbable given enough species in the sample count, but it would be far less optimal in general than simply calling some species relatively more fit than others. Thus, the ancestral species of the Golden Point instating species must be notably fit at surviving the extinction event. And this fitness is a function of two things: the ability to survive the waves of cosmic devastation themselves, and then to outcompete the inevitable slew of other species trying exponentially to repopulate the resulting barren world [16]. The clades which are best at this are more likely to make up a higher proportion of the biosphere millions of years later than those which are not, which either become less and less numerous, or go extinct outright. Then, once this set of clades in the full aftermath are to recover the world, we the humans are, by the First Law, more likely to be born in dominant clades rather than smaller and less successful clades, otherwise we would not have access to the same extent of resources and potential evolution to

the same degree. Hence, humanity is built up of placental mammals and not marsupials or monotremes. Therefore, now if we substitute ourselves back into this hypothetical, in the scope of representing that Golden Point instating species, we can say that our ancestor which directly survived the K-Pg extinction is much more fit to do so than many other groups of animals, and in fact is perhaps even part of a clade which does this better than any single other clade, that clade being the placental mammals.

This being said, it is never yielded through Aurspectative Reasoning how fit our K-Pg ancestral species was individually regarding surviving the extinction event and then repopulating the world better than their competitors, but it tells us that we, (we in this case meaning placental mammals), were best at the optimization combination of both properties together. We may have been worse at one individual one than other species, but together, we were the best there is at mastering both of them together, at least in the scope of all mammals. Whatever the optimal mammal is to survive the K-Pg extinction, and then outcompete any other mammal in its aftermath, it seems as though placental mammals are that mammal. Therefore, even when the Dark Era was still ongoing, and we were making specific evolutionary differences from other mammals, and already it is as if our fate was preparing us for the inevitable extinction event. And when it came, we survived better than any other mammal, even though they arguably shared many of the evolutionary powers of the mind we had unlocked. Simply the placenta, or whatever primarily differentiated us from them, was what made all the difference. By the Third Law alone, we can argue that we would likely not reach the Golden Point without the placenta, but now using this more specific logic, we can also argue that we likely would not have survived the K-Pg extinction without the placenta, or at least if we did, we would not have risen to our dominance of the biosphere, and instead would have gone the way of the marsupials and monotremes under a multituberculate¹⁵ dominated Earth or what

¹⁵The fate of the multituberculates is an especially intriguing query to think about. Going off the general concept of the mass extinction being more or less categorically composed of *the blast* and *the sifting* [16], as I like to phrase them, perhaps

have you [17]. So there is a further connection between arboreal lifestyles, Hațeg Island, and now the placenta itself during the events of the K-Pg extinction event. All three were, for whatever reason, absolutely vital for us getting ahead in the coming Cenozoic rainforest niches.

And of course there is the added component that it was spring when the asteroid struck Earth. Perhaps this was directly related to the function that weeded out non-placental mammals worldwide. Note how no mammal without a placenta is currently native to the northern hemisphere with the exception of the Virginia opossum, which themselves are recently southern hemisphere descendent. Also note how the northern hemisphere takes up the majority of Earth's land, hence connecting it to feasible resources. It is as if Auruspectation's mathematical logic regarding this particular situation makes it so that there is a very specific function which yields the dominance of a clade during the Cenozoic, that being the placenta, and then a statistical limiting factor to wipe out all

there were Late Cretaceous multituberculates who would have been "skilled" at surviving the sifting, however all such multituberculates were already wiped out by the blast. Thus, the ones that had to face the sifting were not capable of surviving through it. I even want to go as far as formally hypothesizing that the presence of placental mammals on Earth during the sifting period was the straw that broke the camel's back so to speak in the scope of multituberculate survival. This is highly un-intuitive in the face of the Dark Era hypothesis, which suggests that certain groups of Mesozoic non-avian theropods were disastrously efficient at putting evolutionary pressure on early mammals. If those theropods, who are literally built for pressure by Auruspective Logic, could not wipe out the multituberculates, then how on Earth could pressure from the placental mammals? Perhaps this should serve as a tale of just how delicate the balance of life was during the sifting. One prior sifting episode – being that of the Permian-Triassic Mass Extinction, was arguably the exact circumstance needed to generate the theropods in the first place, which is striking – and now with the K-Pg equivalent, the entire fate of the multituberculate clade hung in the balance of relatively minute events during the sifting. That may just be how things go when population is reduced so severely, and the environmental changes take place to match. There may very well be many different factors which led to the extinction of the multituberculates once and for all, but it perhaps seems as though the simple evolution of the placenta, which in its mysterious ways would one day bring about the Golden Point, was also something of the ultimate disaster scenario for the post K-Pg multituberculates.

animals not aligning with this norm. If the falloff between fitness of K-Pg survivors and victims was not this high, then threatening theropods would have a chance of slipping through, thus failing the Miracle and launching a global Imperfection Magnitude Drop. So, mathematically, we would expect to live in a world which had a very high contrast K-Pg extinction fitness, and it can perhaps be seen that having a placenta represented the positive peak of said fitness contrast. Essentially what this did was wipe all threatening-theropods and non-eutherian mammals from the northern hemisphere, which allowed our primate ancestors on (hypothetically) Hațeg Island, to thrive quickly and efficiently in the arms race to come to repopulate the Earth. Perhaps the island ecosystem allowed us to build our numbers with relative ease, while back on the mainland no single species could do the same¹⁶ – or maybe there was some other advantage entirely. But whatever happened, without the placenta this advantage would be absent, and without Hațeg Island this advantage would be absent. Thus, by the First Law, both happened in our history.

¹⁶One other quick topic of interest that would be great to ponder is how primates actually got off of Hațeg Island once the Cenozoic was sufficiently underway. As we can see, there are now primates worldwide, so we must ask when this transformation occurred. There are two potential scenarios here: 1) the primates took advantage of oceanic dispersal, as Madagascar lemurs did millions of years later, or 2) the primates developed on Hațeg Island until sea levels were low enough for a land bridge to form, which we logically know must have happened at some point, because Hațeg Island is now integrated into mainland Europe. We may very well expect to find a significant dip in global sea levels during the Paleocene at some point when checking data [18], which we can clearly see happened 58 million years ago, at least as one example. This also, by the way, serves as the very latest possible timecode for the dispersal. By Auruspective Reasoning, we would expect such a sharp dip to be the cause if that was not the general tendency of the time, (as in the world had other reasons / was statistically more expected) to maintain a general consistency of a different sort. However, we would instead expect a more gradual lowering of sea levels to be the cause if the statistical patterns driving Earth at the time could afford it, which implies generally less pressure from other reasons. In this scenario, the dip would more likely be located at 63 Mya, however this feels less valid to my eye because it occurred so shortly after the extinction, which runs the risk of defeating the purpose of Hațeg Island's delicate isolation in the first place. Perhaps both events were contributors of different sorts.

One other quick point of interest is to note that it is technically possible to derive the boreal spring asteroid strike using Auruspective Reasoning alone – however this is in fact a quite difficult theorem to arrive at, to the point where it never even crossed my mind until after I heard of the breakthrough shown with fossil analysis. With that disclaimer handled, let us begin.

The first corollary we need to grapple with is a form of that crucial implication of the Third Law of Auruspectation: the set of events which triggered on that notorious day, 66 million years ago, occurred to the most optimal degree as we can imagine. Any single hypothetical edit we imagine making will prove ill, or even disastrous, for our ancestors and the Golden Point. Thus, we must, under the finite limitations of the human mind, picture this event as the absolute most optimal process and outcome we can imagine.

If we assume the purpose of this proof is to determine the season of the asteroid strike, then the simple question to answer becomes *which season would have been most optimal in favor of the Golden Point?* To get this, we need to determine which specific property this most optimal scenario would have held. I would argue that at least a major player in this most optimal scenario would be some sort of optimization between the highest achievable contrast in the survival spectrum from placental mammals through the rest of the amniotes, with the highest achievable probability of such a scenario playing out. This means we are looking for a scenario which is straightforward and very efficient, but also has that property of extreme survivability contrast among the amniotes.

Thus, if we imagine translating the human ancestral special to the opposite hemisphere from wherever it actually existed, then we may postulate that the new most optimal season would be the inverse season in the year. This is so that the human ancestral species maintains whatever most optimal season they would need to survive the extinction event¹⁷.

¹⁷This does in fact assume that the most impactful aspect of the season is how it directly affects the local environment of our ancestor, rather than a whole multitude of other effects ranging from the jet-stream to the tides, and so on. All hope is not lost, however, because we can come near to negating these effects by simply imagining inverting the properties of the hemispheres,

Therefore, once this most optimal season is found, we can translate it to its hemisphere equivalent based on whichever season it was in the location of the human ancestor. It should also be noted that the extinction should be fully capable of wiping out the non-avian theropods regardless of the season, because after all it will be the opposite season to the human ancestral observed one in the opposite hemisphere. Perhaps, then, the season is more helpful for the sake of clearing the mammalian competition for the later Cenozoic. This means, then, that whichever season best heightens the contrast of placental mammal survival to outside amniote survival is most optimal for their location. It seems that the most severe season would likely be spring, because it would catch the biosphere at its most vulnerable – coming out of winter. If the extinction had triggered during autumn instead in the location of the human ancestor, and many more non-placental amniotes would have an easier time of survival, this directly added to the later competition our ancestors would have to face. Even if this is not a most extremely lower contrast than the spring, it is still something, and something enough for it to have higher Golden Probability, since after all it is equal in Outlook Probability of occurrence to any other season, (because every season has the same length). Thus, it is more optimal for things to go this way than not. The severity of each season, by the way, follows that spring would catch the animals at their least access to resources – having been all used up in winter, whereas autumn would catch them with the most resources gathered. Then there is winter, which is somewhere in the middle, and summer, which has a lack of gathered resources like spring, but at least the animals are more fit and healthy then, as they had not just ascended out from winter.

Thus, spring would hypothetically be the most helpful season to our ancestors, and as we have reasoned already that they lived in the northern hemi-

because after all, there is no fundamental up and down in outer space. The point is, it still seems to be a somewhat good practice to value the tangible environment of our ancestor over all else with regard to satisfying this optimal, because all other properties which lie outside this realm may be invented to any state anyway by the remainder of variables and properties in this Earth-system. To use a colloquial term, “wobble room” is applicable here.

sphere with the logic of Hațeg Island, we can argue that spring in the northern hemisphere would have been the most optimal time for the asteroid to strike. On the modern Gregorian calendar, this is between April and June. And because the Third Law tells us to look to the Optimal for deep time answers, it is most likely that this time range in the year represents the historical impact time during which the Mesozoic ended.

It is quite a windy road to show all that, but it is still hypothetically plausible. This does however mean that some of these very specific attributes could be potentially hard to come by without further information, so had we to rely on Aurusperspective Reasoning alone to solve this query in the first place, we might have found ourselves trying to answer very different questions to start with, such as the specifics of the placenta and the late Mesozoic distribution of non-eutherians. Such concepts would feel unrelated at best without Aurusperspective Reasoning, however with it, we might just find ourselves dwelling on them to piece together the grand puzzle.

And this brings us to another, similar hypothetical. What time of day was it when the asteroid struck prehistoric Yucatán? Again, it seems that our best bet in answering this is to first consider the direct, tangible effects on the human ancestor of this time. If we can derive a most optimal time of day for them to thrive in the scope of the amniotic contrast, then applying this time of day to the longitude of Hațeg is a potentially sufficient answer for the broad time of day during which everything occurred, especially with regards to Yucatán itself. We may even be able to determine whether the sun was out or not when and where the actual asteroid strike occurred. Again, it may feel to be a bit of a stretch to ask whether the human ancestral location had the highest effect on the Golden Point, or rather somewhere else – such as perhaps the entirety of North America. Perhaps we determine that it is worth the optimal sacrifice to do more damage to Mesozoic North America at the relative leisure of Asia, or vice versa, which makes the situation on Hațeg Island relatively immaterial. Truly the best way of objectively handling this conflict would be to consider the contrast of the High Contrast Earth Hypothesis itself. Due to the nature

of how extreme the broad contrast is of the world, at least according to the hypothesis, we can roughly determine that when considering two contemporary effects, whichever one has significantly more contrast is likely to shine through to Aurusperspective Reasoning. If some specific system A has a thousand times more contrast per unit “change” than system B, then we may imagine instating a miniature High Contrast Earth Hypothesis for system A relative to system B, which gives it all the same fruits of logic that we can apply to the main theory. In effect, this means that system A will, for sufficiently high contrast, significantly overtake system B once we apply the Third Law itself, because system B will, at least relatively speaking, be something of a constant in our equation. Near-to statistically independent, if you will, with a far lower multiplicative Golden Linkage. Thus, all things considered, if we can identify the situation with the most contrast, then we have found our primary arbiter of the optimality¹⁸. It seems as though this highest contrast award should go to the events on Hațeg Island itself, for a few key reasons. Mainly, the scale of Hațeg Island in just about every conceivable way is extremely lower than the rest of the world. Accidentally killing one human ancestor in the population which should not have been killed will vastly out-do the killing of any organism offshore in terms of the drop in Golden Probability. The population of Hațeg Island is lower, and every member of that population is significantly more important than any member of any other population, in whatever way depending on the species and individual. Moreover, the mainland world may relatively be killed in great sweeps of destruction by the event, because precision, as important as it may be, is significantly less than on Hațeg Island, where every nanosecond decision is crucial to the coming future. As an example, destroying every living thing in a ten mile radius on the mainland is far less risky than doing the same at any point

¹⁸It is also worth noting that due to the funny nature of the High Contrast Earth Hypothesis itself, when we consider two contemporary systems, it is far more likely that one of them has far greater contrast than the other, at least compared to the likelihood of them being similar in magnitude. The reason, well, is because they themselves have significant contrast between them.

on Hațeg Island, assuming of course that the human ancestor was present there at all. Thus, it is clear to see the extreme contrast difference already.

The next natural question regarding this query is to consider what upfront differences there are between the time of day and the time of year during which the asteroid struck, at least in terms of the optimality. One key element is the temporal span of the death toll itself. If we could imagine mirroring the timeline where the asteroid strikes in boreal spring compared to boreal autumn, and then chart where the differences are in the caused deaths, we might notice that there is still a significant contrast in caused death in the time much beyond the original strike. The season will, of course, significantly affect the individuals of the biosphere in the beginning of the event, but as we can clearly see from the multituberculates [17] who survived the blast itself, the selective weeding out of creatures post-devastation also plays a significant role in triggering extinction. And this, of course, happens far after the blast and original extinctions trigger, so we can see that the seasonal element, as significant as it might have been, has a very lingering effect on the specifics of which species die. The time of day, on the other hand, is potentially quite different than this in practice. Relatively speaking, the time of day during which our ancestors dealt with the original blast probably had a significantly less lingering effect on general death. If we again picture the act of mirroring the timeline where it was day versus night when our ancestors coped with the blast, we might see that most of the differences in death between these two scenarios are over with within the first twenty-four hours. If we could imagine running a clock to chart the deaths of every single individual and species due to the extinction event, then the specific deaths which occur, say, several decades after the blast, will be nearly statistically independent to the time of day of the original blast. This is not to say that the time of day has no long-term effect, but instead that whatever deaths do occur in the significant future after the blast will not have been directly caused by the blast-time-of-day, but instead by other, more overbearing circumstances on them. To put this in a calculus kind of mindset, this is arguing that the “derivative” of sorts of the death toll is statistically

independent to the blast-time-of-day beyond a certain temporal threshold, but not exactly the death toll’s overall magnitude itself. And even calling it a derivative is not quite good enough, because that is based on the counting of deaths more so than the specific properties and circumstances of those deaths. Consider this an analogy if anything, then¹⁹.

All of this being said, we can roughly conclude that whatever affects the time of day had on the specific deaths of amniotes triggered within the first twenty-four hours or so, because beyond this point, whoever was going to directly die because of it was already dead. Narrowing this down further, there is something of a crucial prerequisite to the Golden Point on Hațeg Island during the blast itself: any survivors had to be underground. This is simply due to the insane air temperatures reached for a short time in the moments following the impact [19]. Therefore, we can reason that a scenario which makes it easier for the average primate to get below ground will increase the likelihood of their survival, as opposed to a scenario where it is generally more difficult for each one to descend underground. As an added bonus, as well, if their later-competing lineages of mammals find more relative trouble burrowing underground, then this will only further play into the optimality later on, albeit their lack of a placenta will deal them significant damage as is. A great question to ask now is how much of an effect the hour of day or night will have on this ease of burrowing²⁰ for our primate ancestors. This means that the exact circumstances were potentially crucial. We do in fact know that these ancestral creatures were nocturnal, due to the nocturnal bottleneck [7], so a natural key question regards whether or not it would be more helpful for them to have to wake up from sleep in order to evade the blast, (which the anterior earthquakes might have served as a warning for), or be awake already. Also, generally, it seems likely that these proto-primates might be in many different parts of the rainforest at the time, with some high up in the canopy, and others nearer to the ground, so assuming that this distribution is somewhat random and somewhat uniform, it

¹⁹And, perhaps “change” is at least a slightly better term here than “derivative”, to clean this confusion up.

²⁰... or however they may most efficiently evade the heat...

seems unlikely that the same general rule will apply to all of them. Perhaps, if anything, this is a sign that this assumption is invalid, and that there was some mechanism which led many members of the species to be in a relatively safe-zone during impact, giving them a quick route to underground shelter before the oncoming blast front. However, equal in logical form might be to argue that this is evidence for high populations, or whatever mechanism would allow for there to be “just enough” primates near to the forest floor to disallow extinction. Given our current information, both conclusions are equally valid by Aurusperspective Reasoning – so long as they both optimally help the chances of our lineage surviving the mass extinction, which they do. All of this information being configured together, and the question merely asks *what is the most Golden-respective optimal configuration of properties of the ancestral primate node population directly preceding the earthquakes as pertaining to the hour of day/night, and other offshooting factors regarding this configuration?*. If the population was gathered near the forest floor to mate, for instance, then there is a potential explanation, and one which would assume it was night, (unless it was an intermission in this and they were all sleeping). If these primates had a natural urge to go underground upon being near to the ground and feeling threatened, due to the effects of prior evolutionary ecology, then there is another potential factor. Or, perhaps, the earthquakes might have triggered during the night, such that our ancestors were more awake and aware, whereas perhaps they might be less active in the daytime, a feat which could potentially hurt their chances of success in getting underground in time. Then again, perhaps they had a habit of sleeping underground in the first place, so they were already well locationally equipped to survive. And the list of how things might potentially fit together in terms of the logic, of course, continues. The point is, now the most helpful thing to do in order to determine the time of day of impact is to, believe it or not, uncover some of the specific properties of the population of mammals and their respective environment on, (hypothetically), Hațeg Island, and then how these properties played into the moment of the event. The more we discover about this, the more

yield we can expect to have with Aurusperspective Reasoning regarding the time of day, among other things as well.

11 Properties of Dark Era Predation

During the Dark Era, predation upon our mammalian ancestors was extreme. If one imagines the hazard function of death onto the average mammal as being roughly consistent across populations, then for some region of land with area A and mammalian population density ρ , the number of mammals in said region is $A\rho$. Then, if each hunting theropod has a hazard function of making a kill – so this time making a kill rather than being killed themselves like the mammals – of L , then after time t has elapsed, the probability of them engaging in at least one kill is $1 - e^{-Lt}$. Therefore, given a uniform random variable of u , the time of first kill corresponding to that random score is $-\frac{\ln(1-u)}{L}$. Taking the weighted average of all possibilities from here is $\int_0^1 -\frac{\ln(1-u)}{L} du$,

which results simply in $\frac{1}{L}$. This means that the expected time between theropod kills is $\frac{1}{L}$, so therefore after time t has elapsed, approximately Lt mammals have been killed.

This means that for a theropod population density of P , a total of $APLt$ mammals die. Therefore, the likelihood of each individual being the next one to die at the event of each hunt is $\frac{1}{A\rho}$, so after the $APLt$ hunts and kills have elapsed, the probability of each individual mammal dying is $1 - \left(1 - \frac{1}{A\rho}\right)^{APLt}$. This translates to a hazard function of $APL \ln\left(\frac{A\rho}{A\rho-1}\right)$, which has a Laurent Series first term representation of $\frac{P}{\rho}L$, a very close approximation for either very high areas or very high population densities, both of which are easily attainable for some given forest or beyond. Setting this to the mammal hazard function yields $\lambda = \frac{P}{\rho}L$, or perhaps a more notationally intriguing version.

$$\rho\lambda = PL$$

This conclusion is quite fascinating, because it approximates that the population density of a prey animal times the hazard function of it being killed is equivalent to the population density of the predator times the hazard function of it making a kill.

Note, however, that this set of products, both being equal, may still not necessarily serve as a fundamental property of the predator-prey relation by itself. Specifically, if it is imagined that each predator sweeps out a roughly constant rate of area in their search for their prey, given by some $\dot{A}$, then because it has already been determined that the expected time between theropod kills is $\frac{1}{L}$, it would follow that $\frac{1}{L} = \frac{1}{\dot{A}\rho}$, the right side of which measures the expected time between consecutive mammalian deaths to a theropod given the constant area rate of $\dot{A}$ and the typical population density of mammals, ρ . Therefore, $L = \dot{A}\rho$, which is to say the predator's hazard function of successful kills is written as the area it covers in kills times the population density of mammals. Inputting this into the full equation of $\rho\lambda = PL$ then yields, after simplification, $\lambda = \dot{A}P$. Solving for $\dot{A}$, which has already been supposed to be constant, results in the following.

$$\dot{A} = \frac{\lambda}{P} = \frac{L}{\rho}$$

This $\dot{A}$ can be perhaps taken as the natural ability a predator holds in killing its prey, specifically taken from an ecological standpoint. Thus, because such things remain roughly constant over shorter-than-evolutionary time, this fundamental relation may be thought of as a property of the ecology between a predator population with a prey population, such as, for instance, within the Dark Era. Therefore, taking advantage of this relation, it can be realized that the logarithmic derivative of any of these three relations is roughly zero, meaning the following identity is also true.

$$\frac{\lambda}{P} = \frac{L}{\rho} = \frac{d\lambda}{dP} = \frac{dL}{d\rho}$$

This roughly constant value is perhaps a measure of the natural “predation” of a predator-prey relationship.

I would like to further suggest that a certain subclass of hazard function, acting as a kind of “practical

danger” of sorts, may be written with units morts per time. (Derived just through a simple multiplication factor of the mort [20] onto the preexisting hazard function). Therefore, for every predator-prey relation, there is a fundamental constant of their interaction written with units $\frac{[morts] \times [length]^2}{[time] \times [people]}$. Because the difference between animal and person is so vague from a biological perspective, and especially considering how what we refer to as “people” are in fact descended from animals present in the Dark Era, people has been used as a substitution for “organism” in this particular example, especially due to its inherent relation to morts as per its definition. However, one may just as easily use $\frac{[morts] \times [length]^2}{[time] \times [organism]}$ if they believe “people” is not properly descriptive of ecology, (even though it may be argued that whatever essence of humanity exists in the world today was in fact directly present in certain Dark Era mammals, even if in an earlier iteration of itself).

It seems as though, naturally, the human brain is more accustomed to measurement of hazard function, in morts per time, than actual probability of risk itself. This could very well be evolutionarily purposeful to best ensure our survival of the Dark Era itself. But, resultantly, the brain's perceived gauge of its micromorts per hour is more important in the function of defining fear than the micromorts themselves are. For example, from now over the remainder of our lives, we have seemingly endless more micromorts to go – but this does not directly cause fear in the present. Present fear only spikes when the instantaneous time derivative of those micromorts picks up. That is, the product of the mort with our hazard function. Thus, are hazard functions the default measure of happening for the human brain so to speak, evolved first and foremost as our mind's observational reference during the Dark Era, then yielding the rest of the mammalian perceived sensations on top of that fundamental center? (Also note that if this is the case, then these fundamental mental hazard functions would also be perceived multiplicatively, such that ratios between any two hazard functions are more eye-catching [21] than subtractive differences, the reason being low to high hazard function is thought of as more devastating than already

high to slightly higher, even if by the same additive amount. Thus, if we truly want to represent things linearly with this, we should create a hazard function log-scale to go alongside it mathematically).

12 Temporal Third Law

The Third Law of Auruspectation states that, generally speaking, for some lingually stated Attribute holding Outlook Probability p , the Outlook Probability-based likelihood of the Golden Point in the absence of this Attribute, b , compared to the Outlook Probability-based likelihood of the Golden Point in the presence of this Attribute, a , will follow $\frac{b}{a} < \frac{p}{1-p}$. In practice, this makes our lingually stated Attribute superior to all possible alternative Attributes put up against it, however one catch of this methodology is that while there are hypothetically alternative Attributes which do not satisfy this principle, and instead have a ratio $\frac{b}{a}$ which is greater than the cutoff $\frac{p}{1-p}$, we are not going to actually find them using the power of the human brain alone, and frankly any other finite tools of our capacity. They are simply too difficult to come by, because of the vast logic of the High Contrast Earth Hypothesis's yield.

So even if such more-optimal Attributes cannot be described thoroughly using human language alone, (assuming of course that alternative Golden Treks themselves are not present in the declaration of such Attributes, as in *An Attribute which would cause our Golden Trek to rate with higher Optimality*, for instance), we may still hypothetically picture examples of these kinds of things. For instance, what if there was an Attribute, technically present in our Golden Trek, but so specific and so precise, that an alternative Golden Trek for which one of these precise components was ever so slightly shifted would still have roughly the same probability, thus instantly breaking $\frac{b}{a} < \frac{p}{1-p}$ because in that example, $b \approx a$ and $p \approx 0$. An easy example of such a phenomenon is to simply imagine a Golden Trek for which a numerical Attribute property is precisely a numerical value, as in, say two. Two and a half, or even two and one trillionth would then be outside the allowed set.

Therefore, $p \approx 0$, and so if $\frac{b}{a} < \frac{p}{1-p}$ were to be satisfied, then it would have to be the case for $a \gg b$, which is clearly improbable for a close-enough approach of the Attribute away from two. Thus, using this simple logic, no matter how extreme the High Contrast Earth Hypothesis, it is technically impossible for the Third Law's notational form to hold perfectly beyond a certain level of specificity for the central Attribute²¹. This being shown, let us imagine that our Attribute of choice in this application is to be time. This would make it so we knew the time-code of an event so thoroughly, that it was something of an exact value. If we compared this to all the rest of times, clearly our exact value would be insufficient in magnitude of Golden Probability, however we may notice an interesting characteristic once we think to compare this time to something more similar in magnitude: other single moments in time.

Let us imagine the Golden Probability of some event occurring exactly at time T . We will refer to this Golden Probability as g_T . Let it be declared that this time value is the exact time at which the event triggered in our historical Golden Trek. Then, there may also be the Golden Probability, given as g_t , of that same event triggering at time t instead. Unlike the example of comparing g_T to $1 - g_T$, now we can apply the Third Law as usual to the case of g_T compared to g_t . Hypothetically, if $\frac{g_T}{g_t} = 0$, then it would be impossible for us to exist in the Golden Trek abiding by g_T as we do. Therefore, $\frac{g_T}{g_t} > 0$, and so the two Golden Probabilities are relatively finite. As for why we know the Third Law can be applied in particular, is because we know that the time difference of separation, $|T - t|$, is sufficiently distant from T such that the High Contrast Earth Hypothesis should be applicable, assuming that $|T - t|$ itself is not incredibly small²². If that difference is quite large, then sim-

²¹Also note that this is still great news for the sake of the Third Law holding for all practical scenarios. Because such a tiny precision is required to break the Third Law, in order to actually do this, we would either need to disprove the HCEH, or discover an Attribute of our Golden Trek to such precision, that editing it slightly would not stimulate the HCEH's logic. Either way, this task would be extremely difficult, and so the Third Law of Auruspectation is not going to be broken anytime soon

²²And if that difference is quite incredibly small, then sure,

ply the same logic which made the Third Law is applicable: significant difference from the Earth yields significant contrast, and significant contrast yields a lowering in Golden Probability. Thus, if we apply the Third Law in the set of just these two Golden Probabilities, then we get an individual Golden Probability for each of $\frac{g_T}{g_T+g_t}$ and $\frac{g_t}{g_T+g_t}$. Then, from here, we know that $\frac{g_T}{g_T+g_t} \geq \frac{1}{2}$ by that particular phrasing of the Third Law, and so $g_T \geq g_t$.

Now, for a moment, let us imagine that we could identify the exact range of allowed time within T and t . As we have already stated, these time ranges are zero, such that they define a single point, so let us denote to this some value dt . Assuming that this is equivalent for both times, we see that $\frac{g_T}{dt} \geq \frac{g_t}{dt}$. These may be considered almost as derivatives of sorts, because both numerators and both denominators are infinitesimal. Therefore, if $f(t_0, t_1)$ defines the Golden Probability of the event happening between time t_0 and t_1 , then $\frac{g_T}{dt} = \lim_{t_1 \rightarrow T} \frac{df(T, t_1)}{dt}$. Moreover, for any $f(x)$ which abides by a property $\lim_{x \rightarrow a} f(x) = 0$, it will follow that $\lim_{x \rightarrow a} 1 - f(x) = 1$, and so $\lim_{x \rightarrow T} \frac{f'(x)}{1-f(x)} = f'(T)$. Therefore, because $\lim_{t_1 \rightarrow T} f(T, t_1) = g_T = 0$, it will logically follow that $\frac{g_T}{dt} = \lim_{t_1 \rightarrow T} \frac{df(T, t_1)}{dt} = \frac{df(T, T)}{(1-f(T, T))dT}$, and also that $\frac{g_t}{dt} = \lim_{t_1 \rightarrow t} \frac{df(t, t_1)}{dt} = \frac{df(t, t)}{(1-f(t, t))dt}$. We may now recognize these two values of $\frac{df(T, T)}{(1-f(T, T))dT}$ and $\frac{df(t, t)}{(1-f(t, t))dt}$ as simply the “hazard functions” of sorts of the events, at least where the Golden Probability range function $f(t_0, t_1)$ is represented as a stand-in for the Outlook Probability. This being said, the hazard function of a particular set time should never change if we redefine where on the timeline that time is, (by performing a horizontal translation), and so $\frac{df(t, t)}{(1-f(t, t))dt} = \frac{df(x, x)}{(1-f(x, x))dx}$ for any x . Note²³. Thus, $\frac{g_T}{dt} = \frac{df(T, T)}{(1-f(T, T))dT}$ and $\frac{g_t}{dt} = \frac{df(T, t)}{(1-f(T, t))dt}$. Because we already know that $\frac{g_T}{dt} \geq \frac{g_t}{dt}$, this therefore means that

there is in fact a chance that a more optimal infinitesimal time is to be reached, which yields higher Golden Probability than g_T . But this being said, g_T will still be most extremely close to this temporal crest of highest Golden Probability in whatever bell curve of sorts this might graph into in full.

²³At least for this function $f(t_0, t_1)$, where it is defined as measuring Golden Probability between two times

the following is valid.

$$\frac{df(T, T)}{(1-f(T, T))dT} \geq \frac{df(T, t)}{(1-f(T, t))dt}$$

From here, if we define a function $g(t)$ such that $g(t) = f(T, t)$, then we arrive at the following basic identity.

$$\frac{g'(T)}{1-g(T)} \geq \frac{g'(t)}{1-g(t)}$$

This is a special case of the general $\frac{b}{a} < \frac{p}{1-p}$, because for any Golden Probability $g(t)$, it can be seen to hold that $g(t) \leq e^{-\lambda|t-T|}$, where $\lambda = \frac{g'(T)}{1-g(T)} \geq \frac{g'(t)}{1-g(t)} \geq 0$. It may still justifiably be argued that if the time-range is known to be between $t_0 < t < t_1$ over a uniform probability density function of being anywhere throughout, then the net Golden Probability becomes the definite integral of $g(t)$ over such a range, which itself is fully capable of being greater than 50% of the entire area under $g(t)$ given that $g(t) \leq e^{-\lambda|t-T|}$, so in that case, this formulation is less restrictive and therefore is not a special case of the general Third Law following $\frac{b}{a} < \frac{p}{1-p}$. However, there is also an abundance of cases where this temporal Third Law formulation is actually more restrictive than the general case, as well, especially when considering single instances of time again for the base of some event. The ability of this formulation to reference earlier versions of itself, at least in the direction of time T , is quite helpful, because it caps the progression of event placement over time. Whereas the general Third Law can only tell us that an event will have less Golden Probability if we move it, now this formulation tells us by exactly how much²⁴. Using this more specific bound in temporal situations, we may rewrite the Third Law as following.

$$\frac{b}{a} < \frac{p}{1-p} e^{-g'(T)|t-T|}$$

This is so, where, remember, $g(t)$ measures the Golden Probability of an event happening between times T and t . However, of course, this function is not needed in full in order to satisfy the inequality:

²⁴Or, at least the upper bound for how much.

only its constant derivative value at time T is a requirement. We can think of this $g'(T)$ as a derivative, however perhaps the best way to visualize it is as the hazard function in the scope of all Golden Point-abiding scenarios of the event occurring precisely at time T . Moreover, due to the optimal corollary of the Third Law, we can perhaps also visualize this $g'(T)$ value as the Golden Point scenario-abiding hazard function of the event present at the absolute most optimal time for it to trigger. If such a value is not fully known, however, then the next best thing one can do is substitute in the lower bound for what it may possibly be in place of the precise value. This is so its direction of inequality matches that of the full inequality relating things to the $\frac{b}{a}$, meaning the insertion of any value above this known lower bound runs the risk of making the full inequality inaccurate. And of course, just substituting the universal lower bound of $g'(T) = 0$ simply reduces the full formula back to the general Third Law case of $\frac{b}{a} < \frac{p}{1-p}$, that we might use in non-temporally based sessions of logic. Also note that this may be written in terms of Golden Linkage.

One noteworthy scenario here is where p and $1-p$ are equivalent, which will appear, among other places, when one single Attribute is moved backwards or forwards in time without changing its general contents or Outlook Probability in the process. While this is not as general of a case as the full expanse of $\frac{b}{a} < \frac{p}{1-p} e^{-g'(T)|t-T|}$, its L-Threshold formula has an intriguing relation to Hyperbolic functions. Note here that the actual Linkage object being measured is the Golden Linkage of the event/Attribute occurring at time T , (when it did in real life), as opposed to some alternate time t . This will hold so long as there is equal or more Outlook difficulty of completing the event at the alternate time t , but not necessarily for less difficulty.

$$1 \geq L > L_0 = \tanh\left(\frac{g'(T)|t-T|}{2}\right)$$

As for a quick-and-easy application of this formulation in practice, perhaps we are considering what dangers would be present with postponing the death of the non-avian dinosaurs by an additional

ten million years beyond when it happened historically. From the reference of the modern day in our own timeline, this would move the event from 66.043 Ma [22] to 56.043 Ma, which exists in the Paleocene within our own timeline, roughly contemporary with what would have been the Paleocene–Eocene Thermal Maximum [23]. One noteworthy threat that may have been present during this time would be, at least by the logic of this paper, the sustained longevity of Haţeg Island through the extinction of the non-avian theropods. Another ten million years to continue this is another ten million years over which anything could go wrong, instantly befalling an Imperfection Magnitude Drop on the biosphere in the process. One primary component of this danger can perhaps be seen as the danger of any theropods rafting to the island. This will, in the least, serve as a lower bound for the danger. It may be roughly postulated that the hazard function of a clade rafting to a separate location overseas is inversely proportional to the oceanic distance to be covered²⁵ and is proportional to the original biomass of the clade. Perhaps we can use Madagascar as a rough estimate for this, however note that there is a decent chance that the Second Law is not applicable for it. Nevertheless, we will assume it is applicable anyway, because any differences should really only relate to simians, it may be. Assuming that at least four separate rafting endeavors were made to Madagascar [24], over a course of roughly 400 km, means that a good estimate for the constant of proportionality between hazard functions and distance, at least for this generally sized biomass, is 16 kilometers per million years. Applying this²⁶ to the distance

²⁵The logic behind this is that if every element of the biomass has an equal probability of rafting to any point on a circle of some set radius away from the center, then the probability to a time, (hazard function), should be inversely proportional to the circumference of the circle, which itself is proportional to the radius of that circle. Thus, the hazard function of successful travel is inversely proportional, at least roughly, to the distance to be traveled.

²⁶Yes, this assumes the same biomass size between the theropods and the Cenozoic mammals, which in all honesty is quite a stretch, however it will at least be in the same ballpark. And, judging by the success of Mesozoic theropods, if anything they may have *more* biomass than the mammals of today rather than less, making this mammalian value a lower bound – which is exactly what we are aiming for, anyway.

values of Hațeg Island, which is roughly 200 km [12], and we have a resultling (Outlook Probability) hazard function of likely no less than 80 per billion years. Let us give a temporary name to this rafting hazard function of λ_D , for “Departure”. Then, there is the Outlook Probability hazard function of the K-Pg extinction occurring at that time, or more specifically, of an event happening which renders λ_D obsolete. We will call this savior hazard function, some λ_H . Thus, by comparing both hazard functions continuously to find which occurs first, we may generate the Outlook Probability of the K-Pg extinction occurring before the hazardous theropods have time to make it to the island, which in turn is a Prerequisite to the Golden Point, and is hence a multiplicative factor into it. The following is the probability of this win occurring before time t in the future beyond any focus time.

$$B(t) = \int_0^t \lambda_H e^{-\tau(\lambda_H + \lambda_D)} d\tau$$

From here, we may calculate a few things. First of all, if we call the Outlook Probability of the Golden Point given the successful K-Pg extinction as some G , then the probability of the K-Pg extinction happening before time t , and the Golden Point one day resulting from this, is given as $GB(t)$. Then, there is also the Outlook probability of the K-Pg extinction occurring before the dangerous theropods have a chance to make it to the island period, given by $\lim_{t \rightarrow \infty} B(t)$. Therefore, the Outlook probability of the K-Pg extinction occurring before the dangerous theropods can make it to the island, but after time t , is $A(t) = (\lim_{t \rightarrow \infty} B(t)) - B(t)$, which itself is symbolized with an A to mean *after*. If this condition happens, and then the Golden Point someday after it, it would make the Outlook Probability be $GA(t)$. Therefore, the Golden Probability of the successful K-Pg extinction occuring before any dangerous theropod can make it to Hațeg Island, and also before time t , (but after some reference time 0), is $g(t) = \frac{GB(t)}{GB(t) + GA(t)}$. Now, finally, enough information is present to solve for the derivative of this Golden Probability function.

$$g'(0) = \lambda_H + \lambda_D$$

This also gives us a sufficient lower bound, at $g'(0) > \lambda_D$. Therefore, we may argue that historically speaking, $g'(0) > 80/Byr$. All in all, we may arrive at an upper bound for the probability of the Golden Point given alternate times for the K-Pg extinction.

$$\frac{b}{a} < \frac{p}{1-p} e^{-80 \left| \frac{t-T}{Byr} \right|}$$

And of course, for the particular example involving a 10 Myr time shift, the probability of the Golden Point in this alternate scenario can be found to be 55% lower than it would be with the normal Third Law alone. That is about one more Nescecoin than we would have otherwise been able to keep track of. (Also note that this same drop applies in the past direction as well from the historical K-Pg extinction, such that a mid-Campanian occurence of the K-Pg extinction would add no less than approximately one additional Nescecoin to the Golden Trek compared to what we might have observed otherwise). Perhaps, (and yes, this is a bit of a stretch), we can argue that this same logic applies to the dinosaurs never evolving in the first place, as some form of instantaneous evolution and extinction when they first appeared in the Middle Triassic [25]. Just due to this temporal Third Law logic, the extinction of the dinosaurs in the Middle Triassic²⁷, would have added a whopping twenty Nescecoins to our Golden Trek, amounting to a drop of about one million times likelihood of the Golden Point in Outlook Probability. Maybe this is a sign that the delicate rates of the K-Pg extinction cannot be pushed back so far in time when expecting a meaningful answer, or perhaps, on the opposite side of things, this is further evidence of the High Contrast Earth Hypothesis²⁸. Only time will tell which result is valid.

As for another result which is on the more plausible side of things, we may form an argument on what the alternate Nescecoin content of the Golden Trek might

²⁷Assuming that some other clade occupying their Aursupervective niche did not evolve in their place

²⁸Combined with the previous calculation of the probability of us existing without dinosaurs, the total combination is that our timeline has sixty million times more Outlook Probability of the Golden Point as opposed to one absent of dinosaurs altogether.

be if our ancestors never took aid in Hațeg Island, or something else of the sort²⁹. The direct pressure from theropods to drive mammalian evolution would have ended whether we arrived on Hațeg Island or if the K-Pg extinction event triggered, but interestingly, by the logic of the Temporal Third Law, we can show that had our original arrival on Hațeg Island been swapped out for the K-Pg extinction event in time, our chances of the Golden Point would still lower, and we would pick up Nescecoins accordingly. The best argument which has been made for the timespan of Hațeg Island places its origin in the Cenomanian [12], which means our ancestral lineage would not have been able to take refuge on the island until the beginning of the Late Cretaceous at the very earliest. That bounds our (Mesozoic) time spent on the island to no more than 34.5 Myr. Now, given the coefficient of $e^{-g(T)|t-T|}$ through the Temporal Third Law, this addition would be somewhere between 0 and 4 Nescecoins, depending on when our ancestors originally arrived on Hațeg Island, (which translates to somewhere between 0% and 94% of a lowering of the probability of the Golden Point). If we call the time before present at which our ancestors originally reached Hațeg Island as some t_0 , then by the Temporal Third Law, pushing the K-Pg extinction event back to precisely this moment in deep time will trigger an Outlook Probability of the Golden Point ratio of $\frac{b}{a} < \frac{p}{1-p} e^{-\frac{80}{B_{yr}} |t_0 - 66 Myr|}$. As for tackling that p component, we must consider exactly what two scenarios are being considered, here. One scenario involves human ancestors becoming free of dangerous theropods at exactly time t_0 , but without the aid of an extinction event. The alternate scenario, then, would be where the extinction event does indeed occur at this time. Obviously there is a full infinite stream of other possible scenarios which would occur as well at this time, that do not fall into either of these categories, and hence we can simply disclude such alternate outcomes from our set of analysis. Therefore, the total probability to notice, here, is simply the sums of the individual probabilities of these two scenarios. In order to run the first scenario,

the first thing that must happen is for the human ancestors to arrive on Hațeg Island. The probability of this happening precisely at time t_0 is simply the (Outlook) hazard function of such a happening, some λ_O , (for oceanic dispersal), times the infinitesimal time span to consider, dt , together making $\lambda_O dt$. Note that this approximation for probability given a hazard function only truly holds for an infinitesimal dt , due to the nature of the Maclaurin series of the transformation from hazard functions to probabilities, however it will not hold for any multiplicative timespan factor. Then, once this event happens, the next key requirement is for the island to be free of dangerous theropods at the time of arrival. Assuming that Hațeg Island originated at approximately $\approx 100.5 Myr$, by time t_0 , the probability of dangerous theropods being present there already is, by the nature of our already-calculated hazard function, $e^{-\frac{80}{B_{yr}} \times (100.5 Myr - t_0)}$. Because one of the conditions to this scenario is that we are to arrive at a location free of dangerous theropods, this therefore becomes a required trait. This sets the probability of the Golden Point by virtue of the Hațeg Island scenario at time t_0 to the following.

$$\lambda_O dt e^{-\frac{80}{B_{yr}} \times (100.5 Myr - t_0)}$$

And then the alternate scenario, involving the extinction of the threatening theropods but *not* the extinction of the eutherian mammals, is as follows.

$$\lambda_H dt$$

Thus, p is found.

$$p = \frac{\lambda_O dt e^{-\frac{80}{B_{yr}} \times (100.5 Myr - t_0)}}{\lambda_H dt + \lambda_O dt e^{-\frac{80}{B_{yr}} \times (100.5 Myr - t_0)}}$$

Together with the Temporal Third Law formula, this yields the upper bound of the Outlook Probability of the Golden Point in the face of the K-Pg extinction event occurring at time t_0 rather than the journey to Hațeg Island can be found.

$$\frac{b}{a} < \frac{\lambda_O}{\lambda_H} e^{-\frac{80}{B_{yr}} \times (34.5 Myr)}$$

²⁹ Assuming we actually did take aid in Hațeg Island in real life, mind you

And this may of course be simplified further.

$$\frac{b}{a} < \frac{\lambda_O}{16 \lambda_H}$$

One will notice that in the case of both of these Goodness values of a and b , the likelihood of the Golden Point is being measured directly before the arrival at Hațeg Island, or the occurrence of the K-Pg extinction, exactly at time t_0 . Therefore, directly after time t_0 , the probability of the Golden Point will increase by a factor of $\frac{1}{\lambda_O dt}$ in the case of Hațeg Island arrival, and $\frac{1}{\lambda_H dt}$ in the case of the happening of the extinction. Therefore, we may reassign both variables to hold their values directly *after* the fact of each of their happenings is completed instead of before.

$$\frac{b}{a} < \frac{1}{16}$$

This is the upper bound on the Outlook Probability of the Golden Point where the K-Pg extinction is pushed back to just before our ancestors would have landed at Hațeg Island. As we have reasoned earlier, the pressure from theropods to trigger specific leaps in evolution were already behind us at the point of Hațeg Island, so truly the only thing keeping us from the K-Pg extinction, at least in terms of what is most optimal, is whatever it is that Hațeg Island aided our Golden Trek with. Thus, even if the Mesozoic system had continued beyond this point t_0 , except without our ancestors being present on Hațeg Island or something of the optimal like, and whatever it is that Hațeg Island contributed to our Golden Trek could not be paralleled without generating an entirely new and fanciful happening over that time. That is, the theropod threats onto us were done – as in they had done all they needed to do, so any further threats like this would only hurt the Golden Trek’s efficiency, and thus the Outlook Probability of the Golden Point. The proof of this is the very fact that, historically, these threats did indeed come to an end at time t_0 , at least according to the logic of the Third Law, and the notions of the Hațeg primate hypothesis.

This logic is helpful, because it now allows us to extend the reach of the probabilistic argument of a

pushed-back K-Pg extinction to the domain of Hațeg Island itself: that is, without Hațeg Island or anything doing the same general job, we can say that the Outlook Probability of the Golden Point would drop by over sixteen times, hence bounding Hațeg Island’s Golden Linkage at 88% from below.

And it is another helpful realization regarding where this probabilistic backlash from an absent Hațeg Island would have actually struck the Golden Trek. If the Mesozoic had indeed ended directly at that t_0 time when we arrived at Hațeg Island, then logically, all of that backlash would only have one place to go: the Cenozoic. That is, the Cenozoic would have begun with no more than sixteen times less of a chance of successfully reaching the Golden Point than happened in the real world. This is another helpful logical argument, because it shows specifically how, mathematically speaking, we would have started out the Cenozoic at a relatively huge disadvantage given the absence of Hațeg Island at the Mesozoic’s end.

Lastly for the logic of a Hațeg Island-absent Golden Trek, we may consider the question of what the probability of the Golden Point might be if the K-Pg extinction happened at exactly the same time, but instead we had never made it to Hațeg Island, or perhaps equivalently, we were descendent from a eutherian clade not present on Hațeg Island. Using the corollary that the most optimal time to place the K-Pg extinction in the absence of Hațeg Island would be more or less around the proximity of t_0 , if we added an additional $t_0 - 66 Myr$ to that time to make it align with the real-life time, then the Outlook Probability of the Golden Point would have to drop accordingly. In fact, if we consider the Outlook hazard function of our lineage going extinct under the conditions of the dangerous theropods as some λ_D , which contains any and all factors during that period which would not have been present on Hațeg Island, but *would* have been elsewhere, then the relative probability of survival on the mainland during that time would have been $e^{-\lambda_D t}$. Thus, truly the probability of the Golden Point without Hațeg Island would have been capped at even lower than one sixteenth the alternative.

$$\frac{b}{a} < \frac{e^{-\lambda_D t}}{16}$$

Lastly, we can perhaps relate this theorem with the chances of non-human animals one day evolving to be more humanlike. For instance, if we picture some hypothetical sister clade which diverged from us directly at time t_0 in the past, and stayed on the mainland while we potentially went to Hațeg Island, then at the beginning of the Cenozoic, their chances of evolving toward humanity is no more than the preceding inequality. And because we observe such clades through a more Outlook Probability lens than a Golden Probability lens in many cases, due simply to the Second Law of Auruspectation, then they will perhaps have even less luck along the way from our perspective than we might have had at the beginning of the Cenozoic in a situation absent of Hațeg Island.

The moral of the story is that while, sure, these concepts, at least when acceptably reliable, do not add an awful lot of additional Nescecoins to the comprehended picture, especially compared to the High Contrast Earth Hypothesis, it is still worth mentioning and explaining how the logic of the HCEH as it interacts with the First Law of Auruspectation is fully capable of bringing us to this further addition on the Third Law in many respects. If ever you are in a situation where you need that slightest bit more bound compression, then these formulas may be of use.

13 Size Constraints of the Chicxulub Impactor

It may be noted that the distribution of crater diameters through frequency on the surface of the Earth is an inverse cubic relation [26], such that a crater with twice the diameter of some other, thus being formed by an impactor with approximately eight times the mass, should be expected to be eight times less frequent. This means that an asteroid with precisely a radius of r should have an infinitesimal hazard function of impact upon the Earth of, say, $d\lambda$, which is inversely proportional to the cube of the radius of said asteroid. This may be written as $d\lambda \propto r^{-3}$, or, because proportionality maintains through the multiplication of any constant, $\frac{d\lambda}{dr} \propto r^{-3}$. This is a

much more helpful visualization, because both sides are now nonzero finite. Aside from the intriguing property that this gives rise to, specifically regarding how $\lambda_r(r) \propto r^{-2}$, where $\lambda_r(r)$ is the hazard function of any impactor above radius r striking the Earth, it also follows that the probability of an impact accelerates upwards as the size decreases, meaning by the Third Law of Auruspectation, if the death of the dinosaurs were to be caused by a smaller asteroid, there would also have to be a fall in the probability of success of such a mission.

If one imagines shooting a random asteroid at the Earth to take the form of the non-avian dinosaur killing impactor, then the Outlook Probability of such an asteroid having some size which is a linear scale factor s of the one which wiped out the dinosaurs follows the distribution s^{-3} , because such a thing represents a single moment of the hazard function, just now in the form of a probability density function. Going along with this, there must also be some new probability of the Golden Point, which is of ratio $q(s)$ to the probability which actually occurred. This means that if the probability of the Golden Point following our particular asteroid is a , then in some alternate scenario involving a size of s will have a Golden Point probability of $a q(s)$. Through substitution, this means that the total probability of the Golden Point directly corresponding to an asteroid of size s is proportional to $q(s) s^{-3}$. By the Third Law, this must be less than our own scenario, with $s = 1$, and thus $q(s) s^{-3} < 1$. In other words, $q(s) < s^3$.

That is in fact the fundamental relation, which states that any impactor with a scale factor of s times the one which triggered the K-Pg extinction event must decrease the afterwards probability of the Golden Point by at least the cube of its scale factor. In terms of mass, which is perhaps more intuitive, if the imaginary impactor has a mass of m , whereas the real-life impactor which ended the actual Cretaceous has a mass of m_H , then $\frac{b}{a} < \frac{m}{m_H}$, where b is the probability of the Golden Point in the event of that impact.

As for if the impactor was larger, then the decrease in Goodness could simply be expressed with the rarity of the impactor itself. This makes m_H , then $\frac{b}{a} < \frac{m_H}{M}$. Putting these two formulations together

can perhaps be done with m , then $\frac{a}{b} > e^{\left|\ln\left(\frac{m}{m_H}\right)\right|}$ or perhaps b can be emphasized with the following.

$$b < e^{-\left|\ln\left(\frac{m}{m_H}\right)\right|} a$$

14 Interstellar Panspermia

What if life on Earth originated amongst the stars? Specifically, how would the world be affected if the earliest microbes of Earthly life arrived here from an asteroid or comet with interstellar origins [27]?

While Aurspective Reasoning cannot currently help explain if panspermia is historically accurate, it can yield some fascinating logical conclusions already given the theory that panspermia happened.

To start, the likelihood of life jumping between Solar Systems is clearly very far-fetched, but certain ideas can, under the assumption of the Rare Earth Hypothesis, be more or less weeded out. For one, if life is rare, then it should be expected not to dwell in interstellar and even interplanetary space without some protection, like an asteroid surface to deflect radiation [28]. Such things significantly decrease life's upper bound of commonness beyond what it might have been already. From here, a big question arises given the tenets of Aurspective Reasoning.

At this moment, we must imagine the path that a hypothetical space-rock carrying life might take through space on its way to the Earth. Say it originates from some primordial-biological planet, or for that matter some primordial-biological environment, because even calling it a planet is one assumption too many. That environment shall be entitled The Priben as a portmanteau of sorts. Now, the Priben, at the moment of the event which sent life away from it and towards our own Solar System, was clearly some distance away from the sun. (Which will be a placeholder for the Earth, as we cannot prove life went directly from the Priben to the Earth). This is when things get interesting, because the closer the Priben is to the sun, the more likely it is for life to catch hold of a nourishing Solar System body, exactly like a soccer/football player trying to score a goal. The closer the goal is, the more likely the win,

and hence shooting from nearby is preferable in almost all cases. This function is roughly an inverse square relation, because of the surface area of some imaginary sphere centered at the Priben and passing through the sun. This tells us that, at least according to this property alone, a Priben which is twice as close to the sun has four times the Outlook Probability of accomplishing the journey. Then, if the eventual arrival to Earth is deemed a Prerequisite, which the Third Law would most definitely suggest, this means that the Golden Point increases by a factor of the square of the distance ratio as the Priben approaches Earth, by this property. But in fact, there are other properties which skew things even more. Take, for instance, the probability of survival. If we imagine a roughly constant hazard function of death for the microbes as they endure the nightmares of interstellar space, then the probability of survival decreases exponentially as according to the time that must be spent in those conditions, or, with constant speed, the distance to be traveled, (which itself seems to be highly statistically difficult to accomplish over such a distance [29]). That is, twice the distance will square the standing survival probability as related to this property. The combined Golden Point probability so far is the product of both of these factors.

But things get interesting when the next key factor is considered, and that is the difficulty of nearness. If every interstellar object could be assigned a value charting its forever closest approach distance to the sun, and plotted in a distribution accordingly, it would be noted that the vast majority of them had quite large distance values, whereas only an increasingly smaller fraction would ever pass remotely close. The closer the distance to the sun, the less likely it would be for a given interstellar object to ever match that distance. This is precisely where the resulting conclusion becomes visible, because notice how this property actually favors far distances, whereas the previous two properties do the opposite, and favor close distances. This is an abundant sign of an optimization query. If all probabilities could be multiplicatively combined, then one would notice a crest of sorts for the Golden Point probability at a certain distance away from the sun for the Priben. In this case, where the Outlook Probability of a distance is

already factored into the Golden Point probability, the remaining Outlook Probability can be taken as uniform, and so the Golden Probability itself will be proportional to the probability of the Golden Point through each potential distance of closest approach to the Priben. Hence, by the First Law, taking a random sample from the set of all possible Golden Probabilities in the distribution should yield a result very close to the crest, especially considering the likely extremely low standard deviation associated with such a graph. (As is typical of practically all graphs of this nature, as according to the High Contrast Earth Hypothesis).

This directly means that the measure of the optimal distance away from the sun for the Priben to both arrive at and help trigger the Golden Point from is a very good approximation for the actual historical distance over which panspermia occurred, assuming it did at all. This also means that panspermia would have been noteworthily short of a distance – maybe even below a few thousand astronomical units³⁰! After the transfer, of course, the Priben would then continue on its interstellar path, never to be seen by us again. (Also note that this hypothetical optimal distance is probably a function of many more variables than have been stated here – for instance something as seemingly obscure as the average density of interstellar space could be the key to success in this improbable endeavor).

There are a few other logical conclusions that can follow this major one. The first few regard the Second

Law. Perhaps the most striking is that, statistically, the Priben must now be utterly devoid of life, and likely has been that way for billions of years. Finding the rough analog to the danger of an external Departure Point on a cosmic scale, life is, in general, not likely to survive very long in the bath of radiation and rapidfire of threats that exist continuously in our galaxy and universe. Because it can safely be concluded that the Priben, as of a few million years after the transfer, had become permanently statistically independent from our Golden Point, this means that the Outlook Probability of it surviving matches the Golden Probability of it surviving. At this point in time it is likely not causally connected to us in anything under dozens upon dozens of millennia, so this statistically independent nature, especially now, is rather obvious. Hence, the likelihood of it still containing but one single cell in the modern day is next to nothing. This means that future Golden Point life-forms of our descent may never identify the Priben at all in our galaxy, amongst the swaths of burnt and destroyed worlds. However, even if, hypothetically speaking, the Priben biosphere were somehow still alive, identifying it would be difficult anyway because of its notoriously low biosignature.

Moving back in time to the hypothetical pre-panspermia environment, when Optimisation³¹ was still covering the Priben’s survival – there is something to be said for potentially alien biosignatures there, but it can nonetheless be logically concluded that molecular oxygen never was present there. If life on Earth did indeed descend from this hypothetical environment, then the physically transferred life would have been in the form of a colony, and thus is very likely pre-LUCA, (the last universal common ancestor [30]) in type. (On the off chance that this is not the case, then it would mean that both bacterial and archaea life were aboard the transfer-object, which squares the survival likelihood – or in other words seems too far-fetched for Optimisation to choose it anyway unless there’s some phantom Prerequisite in the mix there. All around unlikely). Any life which is this ancient holds some key properties of difference from you or I. Perhaps the most noticeable are the

³⁰The reason being the danger of death grows so quickly for the microbes, and the probability of successful aim decays so significantly for higher distances, that the stellar transfer should not, it seems, be anywhere near the “usual” distances between stars, such as for us and Proxima Centauri. At that distance, the derivative of Golden Probability to stellar distance appears to be negative because of the nature of this stacked likelihood of ill fate for such a far placement of things. On the opposite side of things, if the distance were much closer, at say slightly extra-Neptunian distance, and the opposite effect would be true, with a likely positive derivative, due to the extreme improbability of such a transfer lining up so perfectly. Thus, the distance should be well under a lightyear, but well over one hundred astronomical units. Beyond these bounds, we can only approximate the answer, (which itself may be possible with the right substitutions into the proportionality arguments already listed here, and more).

³¹See the Optimisation section for detail on this word.

lack of photosynthesis and a need for oxygen. In fact, they would have found oxygen toxic [31]. Because oxygen on Earth, and our ability to breathe it, was a result of miraculous happenings which spawned photosynthesis and eukaryotic life, by the Rare Earth Hypothesis it can be seen that such happenings were beyond unlikely to trigger successfully. Even if oxygen in the air supply of the Priben were statistically independent to the Golden Point, and still the likelihood of it being the case would be practically nothing by the Second Law and the Rare Earth Hypothesis. In essence, for there to be oxygen there, something biological would have to be producing it, which itself is a statistical near-impossibility. But, because of the toxicity to the life of the Priben, it is probably the case that the Golden Linkage is actually negative, and so the likelihood of there being even a hint of oxygen is less still. In the end, the Priben, even at its most nourishing to our ancestors, would be inhospitable to modern humans without spacesuits.

Then, at some point, one single colony from the Priben arrived at Earth, marking the origin of life here. Whether panspermia is historical or not, the origin of life on Earth was probably during the Hadean instead of the Archean according to my logic. The reason for this is something of a trend to be noted in clade origination, which is that the last common ancestor of a clade is more often than not older than the average consensus among scientists. In other words, we have a tendency to underestimate the age of clades. Clades themselves cannot become widespread enough for the chance of fossilization until often millions of years beyond their origination, so when the first hint of their success is immortalized through fossils, they had already been through a period of relative obscurity for quite some time. Obviously the degree of underestimation decreases with younger clades, as fossils are more abundant, and finer stratigraphy is better understood, but this general razor still, on average, probably holds.

The last major logical conclusion that would follow the event of panspermia in Earth's past, is what it would mean for the alternative case. In essence, if panspermia were historically accurate, then abiogenesis on Earth would logically have to be an Imperfection Magnitude Drop. This can easily be demon-

strated just with the High Contrast Earth Hypothesis itself, but as it turns out, just using the simple inequality of the Third Law and or the mediocrity principle, the same Imperfection Magnitude Drop result may still be attained, (which shows just how severe it really is). If the likelihood of abiogenesis in the Priben is written as a_0 , the likelihood of successful travel to Earth is written as b_0 , and any leftover probability required for the Golden Point is written as c_0 , then $a_0 b_0 c_0$ is the net probability of the Golden Point with respect to the origin of life in the Priben. Then, if the likelihood of abiogenesis on Earth is some a_1 , the likelihood of Earth life arriving on Earth is b_1 , and the leftover probability of the Golden Point from there is c_1 , it would follow that the net Golden Point probability with respect to the origin of abiogenic life on Earth is $a_1 b_1 c_1$. Note how each variable has an analog in the alternate scenario, and also note that $b_1 = 1$ clearly, so the probability of the fully Earthen Golden Point is equivalently $a_1 c_1$. From here it follows using the specific Golden Point-scenario-centric aspect of the Third Law that $a_0 b_0 c_0 > a_1 c_1$ so long as panspermia is historically accurate. Solving for $\frac{a_1}{a_0}$ from here yields the relation.

$$\frac{a_1}{a_0} < \frac{c_0}{c_1} b_0$$

Because the Priben is likely significantly older than the Earth, it thus has that much more time for an external Departure Point to potentially trigger, and hence it is likely that $c_0 < c_1$. Beyond the convergence of the two scenarios with the arrival of life on Earth, these probabilities are equivalent, and so any difference is likely located in the Priben's danger of the external Departure Point. Because $c_0 < c_1$, this would mean that $\frac{c_0}{c_1} < 1$, and so $\frac{a_1}{a_0} < \frac{c_0}{c_1} b_0 < b_0$. By the transitive property, then, the resulting property is reached.

$$\frac{a_1}{a_0} < b_0$$

Recall that b_0 is a very small number, because it measures the probability that life from the Priben miraculously travels to Earth unscathed. The Priben must approach close enough to Earth for the exchange, the exchange mustn't miss the Solar System, the life must survive the journey and the impact, and

the list of improbabilities continues. This means that, because $\frac{a_1}{a_0} \approx 0$, a_0 is vastly greater than a_1 , to the point where abiogenesis on Earth would be an utter Imperfection Magnitude Drop, otherwise general Optimisation would not bother with the far-fetched endeavors of the Priben and panspermia in the first place. (Again, remember that this may only hold if panspermia is actually historical).

$$a_0 \gg a_1$$

Lastly, it does not have to be assumed that panspermia sent life directly to Earth. Jupiter, for one, has a surface area of over one hundred times more than Earth, so unless the original arrival of life on Earth is more than one hundred times more beneficial to the Golden Point than life arriving on Jupiter, only to transfer to Earth later, it is actually more likely that panspermia arrived at Jupiter in place of Earth.

15 Interplanetary Panspermia

Noteworthy contrasting from the grand idea of life arriving to Earth from the interstellar heavens as is supposed from the hypothesis of “Panspermia”, there may also exist a smaller scale concept of what happens when life travels between neighboring planets.

In the past, Venus, Earth, and Mars were all habitable, which seems like hardly an accident. If the habitability of our neighboring planets was statistically independent to the Golden Point back then, then by the Second Law, it would follow that the arrival of water on Earth was likely a phenomenon of the entire inner Solar System rather than only here. The converse of this is not necessarily true, so water still could have arrived indiscriminately across the entirety of the inner Solar System regardless, but one specific point of interest is perhaps why we have these neighboring planets in the first place, located as they are and such. If Mars, say, never had liquid water, it can be reasoned by the Third Law that the Golden Probability would decrease, but the question beyond this becomes whether this decrease would be due to difficulty in Mars avoiding the onset of Solar System-wide water movement, or some inherent Linkage between

historical water on Mars, and the Golden Point of Earth.

One way to get to the bottom of this is to try and consider the Solar System as a unit biosphere rather than simply the Earth. There is talk in the scientific community of life on Mars being bad news [32], because it would surely move Great Filters forward in time and make abiogenesis more likely than we thought – but this need not necessarily be the case at all. If we identified life on Mars tomorrow, there are essentially two main causal possibilities. For one, maybe that Martian life independently spawned, with an entirely separate tree of life and an entirely separate abiogenesis. Such a thing would, indirectly if not directly, invalidate the Rare Earth Hypothesis, because rendering such a large chunk of the Golden Trek probably would likely send waves of change throughout the entire thing, taking down the High Contrast Earth Hypothesis and all Auruspective Reasoning. The Rare Earth Hypothesis would become a scientific relic of the past. However, logically speaking, this is probably not the case. And that opens the door to a potential secondary solution to this problem if we ever came across it.

Let us imagine, momentarily, that there is some constant hazard function of life of any type arriving on Mars by asteroid or comet. This transfer could be as simple as a high velocity impact on Earth which shoots at least a few cells in the direction of Mars, allowing them to make it there without dying [33]. This quantity shall be referred to as λ_+ . Then, there is also the hazard function of life on Mars being wiped out to consider. Perhaps there is a planet-wide super-freeze, or the collision with Phobos, among other methods of Mars being utterly sterilized of life. This quantity shall be λ_- . Note the use of the plus and minus subscripts to denote life arriving on Mars, and then exiting Mars through death or some other mechanism which ceases its presence there. All of this being considered, the Outlook Probability of life existing on Mars is precisely as follows.

$$\frac{\lambda_+}{\lambda_+ + \lambda_-}$$

If a microbe existing on Mars is statistically independent to the Golden Point, which it likely is, at

least in this day and age, then the Golden Probability should be equivalent to this by the Second Law, and hence by the First Law, the probability of there being life on Mars with respect to human observation is $\frac{\lambda_+}{\lambda_+ + \lambda_-}$. This same formula can be used for any object in the Solar System besides Mars, from Jupiter to Enceladus to Triton. In the case of Triton, of course, $\lambda_- \gg \lambda_+$, so we are not likely to identify life there. As a matter of fact, this same formula may even be used for some specific plot of land on any cosmic object, or even for some specific clade or set of clades of life in some specific plot of land or set of objects. The point is, the application is wide and creativity-accepting.

Considering the set of all of these objects and more, the probability of life not existing in any of them is some $\prod_{k=1}^N \frac{\lambda_{-(k)}}{\lambda_{+(k)} + \lambda_{-(k)}}$, which notice decreases for a greater and greater N value, the set of all potential objects. This may not tend exactly to zero, and in fact it very easily could not if $\lambda_{-(k)}$ grows enough for increasing k s. But nonetheless, the probability of life existing anywhere outside the direct Earth in our Solar System must be greater than the probability of it existing at any one (non-Earth) location, to the point where it is very feasible that we end up detecting LUCA descended life somewhere in the Solar System. And if this does indeed happen someday, so long as the Rare Earth Hypothesis and the rest of the general conclusions of this paper are accurate, that life will have been descended from Earth life which conveniently was transported there on some natural space debris.

Where $\frac{\lambda_+}{\lambda_+ + \lambda_-}$ gives one the probability of finding life on a planet at any given moment, the probability of it ever existing there throughout the entire Golden Trek is much higher. In fact, there have probably been waves of life transferring to a planet, before a decline, and then followed by further waves. The declines themselves are probably not incredibly sharp, at least not usually, meaning while they have the potential to wipe out huge portions of life on a given planet, they still leave a certain amount alive. This amount may be small, but it is usually something rather than nothing. And once the apocalyptic event is over, that life manages to repopulate the world,

filling in the old environments that the previous generations once occupied. Think mass extinctions, but for microbes. But this is only the first part of the process, because in the midst of all of this growth and decay of life on a planet, there is also a factor of life arriving from other parts of the Solar System, so in the end, if the diversity of life in our Solar System could be played like a timelapse, we would observe a grand machinery of transfer between worlds, each step of the way heightening and purifying evolution through perpetual introduction and removal of different clades into each environment. Perhaps if we could trace our own lineage backwards in time, we would find ourselves on several different planets during different eras, up until we converged onto a single world amongst all the rest of life with LUCA. This phenomenon is less a question of if, and more a question of how fast the time lapse must be sped, considering how the past habitable planets of the Solar System must all, at some point, be enriched with life. This concept gives rise to potentially entirely separate evolutionary competition environments depending on the iteration of transfer, which at its best could be abundantly beneficial to Optimisation's workings.

The era of prokaryotic microbes on Earth was a long one, so they were likely the primary drivers of this conveyor belt of motion throughout the Solar System for quite some time. But at some point, eukaryotes came onto the scene, so they might have had a role in all this as well. If we identified oxygen on Mars or Venus in the past, this would in fact be great evidence that eukaryotes were once present there. Molecular oxygen in and of itself requires a certain set of milestones to achieve within a Golden Trek, and while we know this did indeed work out in our own Golden Trek, that does not clear our neighboring planets of all respective difficulties. Cyanobacteria were essentially the cause of atmospheric oxygen on this planet, so logically they could very efficiently be transported to Venus or Mars at some point or other in the Golden Trek and still hold a satisfactory high Golden Probability. But this means a set of attributes must be satisfied. Not just any cosmic body can be home to cyanobacteria and their characteristic oxygen production – so the set of Prerequisites on Earth which enabled this must have also

been satisfied on any planet which did the same. For instance, the study demonstrating that Earth's rotational speed is key to its oxygenation would have to also be satisfied on some alternate planet, at least if some other more efficient mechanism was not statistically probable and or doable. This of course is only an example meant to hint at this underlying mathematical trend, but regardless if it itself is valid, we would notice that the Outlook Probability of oxygen filling the atmosphere of another world in addition to the Earth at some point or another is noteworthily low. And by the Third Law, this puts the L-Threshold at a very high value, and hence, so long as we had proof of abundant molecular oxygen in some other planet's ancient air supply, the Golden Linkage would be confirmed to be significant. The question then becomes why. The most probable out of the potential solutions to this is that the life there was dependent on that oxygen, and because such a dependence is a telltale sign of eukaryotes, one would be faced with striking evidence that eukaryotes once inhabited the planet in question.

If eukaryotes were indeed found to have inhabited some alternate planet, then the natural next few questions would be how this is useful to the Golden Point, and perhaps whether those eukaryotes were ancestral to humans or not. If they were, then the results could be staggering for the uncovering of new science. And even if they were not, then the results could still potentially be just as staggering, because maybe we would find that life to be ancestral to some other vital eukaryote like fungi, plants, or maybe something even more outstanding, (and controversial for it), like mollusks or arthropods. For all we know, this process, in its final breaths of having noteworthily Golden Linkage, might have triggered the Cambrian explosion, however that is obviously speculative beyond the use of modern science at least. But if a tardigrade can survive the journey from here to Europa, say, then it's not off the table quite yet.

16 Difficulty Dynamics

It can be clearly seen that Golden Probability may be derived from the Goodness in either of two scenarios,

along with the Outlook Probability of the difference between those scenarios.

$$g = \frac{p a}{p a + (1 - p) b}$$

Furthermore, it may be calculated that there is such a thing as "Golden Linkage", formulated as follows.

$$L = \frac{a - b}{a + b}$$

Golden Linkage tells of the value some Attribute holds to the Golden Point, especially including what trouble would be caused in its absence. But, for all its use, Golden Linkage says nothing of the difficulty or easiness of attaining an Attribute. Instead, it merely tells how helpful that actual acquisition would be. Golden Linkage does such a fine job of encompassing both Goodness values in either scenario, that it may, along with the Outlook Probability of the Attribute, fully encapsulate its Golden Probability.

$$g = \frac{(1 + L) p}{2 L p + 1 - L}$$

This is certainly notationally helpful, because it manages to reduce the equation to three variables rather than four with no information lost in the process, but at the same time, it does not have quite the same beauty of form as the original formulation of $g = \frac{p a}{p a + (1 - p) b}$. The reason for this, simply, is because Golden Linkage is of almost another "type" of quantity than Outlook Probability, similar to how forces fall under Newtonian Mechanics and the lagrangian falls under Lagrangian Mechanics. They arise in different systems, and so combining them inevitably leads to messiness. Therefore, this problem may be easily solved if only an analog to Outlook Probability were to be found within the new system. This analog, is what is called "Easiness".

Easiness is written simply as follows, where p represents the Outlook Probability of the Attribute.

$$\xi = \frac{p}{1 - p} = \frac{1}{\frac{1}{p} - 1}$$

Likewise, there is also Difficulty, the reciprocal of Easiness.

$$\xi^{-1} = \frac{1 - p}{p} = \frac{1}{p} - 1$$

The higher the probability of success of accomplishment, the higher the Easiness, and the lower the Difficulty. Easiness is the probability of success divided by the probability of failure, if you will, with Difficulty being the opposite. But the great thing about these quantities is that they are not restricted in the set from zero to one, and thus may scale by any value as according to some variation. For instance, where there is the threat of going outside the range in the act of doubling Outlook Probability, for Easiness and Difficulty there is no such threat. A great starting identity to illustrate the power of these quantities is the realization that the Easiness of doing something is always equal to the Difficulty of not doing it, and vice versa.

With these quantities being known, now the Golden Probability of an Attribute may be written purely in terms of the Golden Linkage and the Easiness of an Attribute.

$$g = \frac{(1+L)\xi}{(1+L)\xi + (1-L)}$$

Note the striking notational similarity to the typical formulation, where the quantity found in the numerator is also additively present in the denominator, along with some other dual quantity.

$$g = \frac{pa}{pa + (1-p)b}$$

A fascinating and helpful property of the duality of Easiness with Golden Linkage should also be noted. A general pattern of optimization is the presence of two “competing” elements of a system, which is a great tool for spotting where and when to use Auruspertive Reasoning itself. This same general idea may be observed in the spectrum of high Golden Linkage to high Easiness. Oftentimes, (at least in the set of all possibilities, making this not necessarily a commentary on Earth’s physical past), Golden Linkage is extremely high at the detriment of general Easiness, and on the flipside of things, Easiness is often quite high at the detriment of passable Golden Linkage. Therefore, these quantities can be seen to, at least in some sense, compete. So, using the same underlying logic as optimization, it can perhaps be seen that in fact the “correct” amount of Golden Linkage

and Difficulty will stand in a sort of middle ground between the extreme points of either. (Correct meaning what is most likely to be observed in the Earth’s past as according with The First Law). Increasing either at the detriment of the other is less likely to be observed than finding some sort of “compromise”, to personify for the sake of analogy. This compromise is what shall be optimal, and thus is what is most likely to be seen pastward.

Additionally, when using Difficulty, written as ξ^{-1} in place of Easiness, this fundamental relation may become the following.

$$g = \frac{(1+L)}{(1+L) + (1-L)\xi^{-1}}$$

This form is perhaps superior in its lessening of total variables visible in the notation, however it suffers for the lack of intuitive optimization visualization.

The next natural step from here is to describe the Third Law using this information. It has already been shown that it may be written in two ways.

$$\frac{b}{a} < \frac{p}{1-p}$$

$$L > 1 - 2p$$

Now, using Golden Linkage and Easiness, we may also add the additional entries to the list as corresponding to Easiness and Difficulty, respectively.

$$L > \frac{1-\xi}{1+\xi}$$

$$L > \frac{\xi^{-1}-1}{\xi^{-1}+1}$$

17 Task-Object

We as a species are often prone to stereotype, oversimplify, and altogether skew our understanding of the universe unnecessarily. That is just what stereotyping is, beyond even the moral argument: it starves one of the purity of their observational reference. More specifically, even if their observation remains untouched, through stereotyping they are cropping

out swaths of their information inflow, which is somewhat depressing in its concealment of the awe and beauty of the universe. But then there is the immortal fact, which is that we all do this in some form or another with our very existence in society. Societal norms and accepted viewpoints influence the raw perception of the outside world for each and every one of us, and the extent of this is, in a word, striking. The process of beating past this blurred vision is just as vital to great strides in science as the forward momentum itself. If we consider the set of all science which has not yet been trodden into, then what makes this set this set is the very fact that no one in history has yet broken the underlying stereotypes of the perceived human experience that need to be broken, hence falling the first domino to spark the chain reaction which widens our perspective. If we had the theory of everything right now, and we compared it to modern science, my bet is that the fundamental differences would fall at the very roots of what we think we know about the world around us.

The point of including all of this is to demonstrate just how flipping one fundamental concept of reality can change everything. Take non-euclidean geometry, which alters Euclid's absolute fundamental postulates like the distance between two points – an entirely new systematic way of thinking may be birthed from the tiniest changes upfront. So then, for the query here – how does one define an object?

It seems as though the classic way of describing an object regards its perceived independence from other objects. There is nothing inherently wrong with this approach in the context of many of the aspects of our society and day-to-day interactions. But is it truly the most fundamental definition of an object in general? For instance, there is also what I call the “task-object”. It is not a more fundamental definition or even a substitute for object, but instead is to be used in specific cases where the concept of object becomes too arbitrary for the conditions.

A task-object is an object defined not by perceived independence from another object, but instead defined by a contribution it pays to the Golden Point. Take, for instance, the Earth. The Earth as an object is easily distinguished from a planet with dif-

ferent continent configurations, atmospheric composition, diameter, and so on, as we can clearly see by calling Alderaan from Star Wars not Earth. But, even though it satisfies many of these properties of difference, Earth 3.9 billion years ago is still called “Earth”. However, a perfect Earth duplicate, if it were to exist in a separate star system, would likely be called “Not Earth”, and so the confusion grows. But the strangest part arises when we consider Earth in the scope of the Golden Trek. If we imagined another Golden Trek which had no “Earth”, then this can be easily achieved by simply taking down the exact set of factors which would lead the informed human to define a planet as “Earth” from the Golden Trek. In its place, of course, a planet very similar to Earth would have to be substituted in place, otherwise an Imperfection Magnitude Drop would trigger. But, technically, it is not Earth, because if we compared it to our world, we would shake our heads and declare that it is not in fact “Earth” at all. This is all well and good for certain purposes of use for the Earth, but it is problematic in others. For instance, if we were defining a brand new Golden Trek, say as a hypothetical separate dice roll within the scope of the First Law, then it would be quite useful to define the analog to Earth, and all the rest of the objects of our Golden Trek that we name Prerequisites or in the least relevant. So, perhaps we might define a “Task-Earth” as the planet on which the Golden Point's makers speciated, or perhaps something even more fundamental in its logical and intuitive nature. (For example, if we find “speciated” itself to be too dependent on objects rather than task-objects, or in other words is too specific to our Golden Trek, then we might find a better substitute, or just establish a more clever Task-Earth definition altogether).

The trouble with translating directly from an object to a task-object, is that any one object could be broken into infinitely many task-objects, and also that any one task-object could be broken into infinitely many objects. That is, the Earth as an object could be assigned many different task-objects which are inclusive of the Earth as an object, like say “a spherical thing composed mostly of iron”, or “a region of space in the goldilocks zone of a star”, and the infinite list continues. Also, if any one of these

task-objects were to be selected, then an infinite number of objects could be imagined which all satisfy the definition of that particular task-object, like naming a Task-Earth of speciation location for the bringers of the Golden Point, and then defining all possible Earth objects which satisfy this Task-Earth. For instance, if there were an alternate reality where say Hawaii did not exist, then while such a planetary body would not technically be the exact same Earth object as our own, it might very well still count as a Task-Earth, and hence for each task-object there are infinitely many qualifying objects. This of course means there should be some subjectivity in the mix of defining and naming task-objects, but perhaps it is subjectivity which may be linguistically systematized.

Of course, as there are task-objects, there may also be task-humans, task-events, etcetera. But once we decide to create this classification system, we can not only see where some of our biases were all along, but we may peer into the concept of the generic Golden Trek with greater accuracy.

18 Optimization

Auruspective Reasoning serves as a vivid mathematical and law-oriented framework, but it still might be equally intriguing to picture it in a different way altogether. That is, what if it could be viewed as a “causal” process instead, similarly to evolution?

Let us imagine that there was a fundamental process which was located at the Golden Point in time, but pointed backwards in its effects rather than forwards. We shall call this imaginary process, “optimization”. Now, optimization’s primary goal is to set up optimal events to happen through the effect of previous events. This perhaps is not a causal relationship in the Copernican sense, but instead just an interesting effect of these concepts, but it is observationally concrete enough that if it is imagined to be true, it serves the purpose of helping one visualize the grand scheme of how and why things happen on Earth, and how all of this plays into Earth beating the seemingly impossible over and over again.

An example will be helpful for this. Take the chem-

ical α -gal, which mammals evolved an ability to produce during the Dark Era. It is helpful toward body cell identification, but this is likely only the start of the applications of it, especially considering how it is particularly optimal as cellular chemicals go. During the Cenozoic, the genes to produce it were lost only in catarrhine primates [34], a clade including humans. Any time the production of a chemical is evolved in a clade, before being lost in a further subclade, itself containing humans, the logic behind Auruspective Reasoning is very likely at work. The scenario where that chemical was never able to be produced in the first place must have lower Golden Probability than our real scenario by the Third Law, and so the likelihood of the Golden Point in the event that we never evolved α -gal, (or something roughly analogous to it in whatever mystical role it plays in the Golden Trek), during the Dark Era would have to drop. Hence, α -gal holds some Golden Linkage. But, later, losing it had to be important as well, because of some primate-type importance factors. Mice which had its production removed from their genetic codes were found to not be able to reproduce in quite the same degree of youth as before. Perhaps this is an insight into why it occurred in the Dark Era, because it aided mammals in reproducing from a young age during the transition to live birth – and just that head start in reproduction throughout life might have been a substantial help to the Golden Point by keeping mammal populations more afloat, and thereby diminishing the Departure Point’s probability by that much more. This particular aid is at this point hypothetical at best, so it is meant to be a hint that something optimal is likely going on behind the scenes here regarding α -gal and the Dark Era.

But, of course, eventually the Dark Era ended and eventually α -gal production had to go. This, again, logically, has to have some sort of optimal nature, whatever it may be, especially considering how it happened. First of all, no other mammal lost α -gal, so that automatically increases the rarity of the situation. Furthermore, this loss was likely due to a virus – and one which possessed high amounts of α -gal. This horrible virus struck our ancestral monkey populations millions of years ago during the Cenozoic, and it had been so deadly, that our best evolution-

ary bet at beating it was to cut α -gal out of our cells altogether, so any cell in our bodies still holding it – which would all be virus cells – could be targeted. This lucky loss of α -gal likely saved our lineage, especially considering the evolutionary weight that loss held. This was a very close call, because it goes without saying that the death of the human lineage at this time would have been devastating, and very potentially even could have been a full Imperfection Magnitude Drop, which would then have made our counteracting α -gal loss technically Miraculous. The virus evolving, perfectly containing α -gal, and then not wiping our race off the planet, all contribute to this generally being a moderately unlikely situation. Hence, its Golden Linkage is probably high.

What follows from this is that the loss of α -gal production is not only important to the Golden Point, but also difficult statistically to occur at all – otherwise the Golden Trek would have just gone down an easier path to rid us of it, which would have had higher Golden Probability. But no such scenario occurred, and thus no such scenario exists. Further evidence for that regards how no other mammal seems to have lost it, and hence there is a clear Golden Probability dependence on all of this going down in the first place. This means that evolution itself was resisting the loss of α -gal in general mammal evolution, probably again relating back to how wildly helpful it would have been during the Dark Era, and thus the very nature of mammalian life. And yet, somehow, we lost it anyway.

After its loss, because of the optimal nature of such things, we must have been aided in some form or another in becoming humans. While, again, the actual areas of aid are at this point speculative, it might very well have had to do with humanity's very long childhoods, an attribute necessary for our large brains and achievements. Whatever the purpose, or purposes plural for that matter, in the end the loss of α -gal production directly led into the next key steps in human evolution. Had it not triggered, these steps would have to be reworked with much lower Golden Probability.

And this is where optimation comes into the picture. While there are perfectly good explanations for why this virus appeared to force our evolution

to remove α -gal production according to the Third Law alone, if we picture the process of optimation as being involved, then now we may simulate a version where optimation actually caused that virus to evolve by facing backwards in time from the Golden Point and triggering the foundational event and allow the Golden Point to happen at all. Relative to the Copernican Principle, this would rightfully seem absurd, so the mindset should instead be to consider it relative to human observation. There, things change, and it becomes no less causal than evolution itself always has been. Think of it not as an actual concrete force, but more of a framework which simulates the happenings in an intuitive sense. Optimation would have triggered the evolution of the virus, and from there it also made us survive the virus, enabling us to move onward through our Golden Trek. Evolution and the rest of the formally causal processes of our reality were technically the ones making things happen, but it's almost as if they were being guided that whole time by the backwards-working optimation. Without the forces of optimation, the virus never would have appeared in the first place – because that would have been unlikely – and for that matter, sooner rather than later the external Departure Point would simply destroy the biosphere, because that too is more likely than not. Without optimation's effects in place, everything would become essentially statistically independent to the Golden Point, the Golden Probability would collapse to match the Outlook Probability, and the Copernican Principle would take over, rendering "Earth" as a lifeless mass of toxic metals and minerals either on fire from the sun's fury, or floating frigidly as a rogue planet between the stars.

Optimation itself can work on lifeless entities like the asteroid which killed the dinosaurs or the drift of continents as well as biological ones like viruses and populations. But other than this difference, it actually has some interesting parallels to evolution by natural selection. While natural selection tends to give organisms more optimal methods of survival and reproduction, optimation tends to give the Golden Trek more optimal methods of setting up the Golden Point. But because events are fundamentally made to influence future events through optimation, it has something of a backwards-facing transfer process, at

least more so than forwards.

As for a more in depth analysis of what was proposed in the introduction, regarding how “the rain optimized to enable the Golden Point, and some particular grass evolved to take advantage of this happening”, the Golden Probability of things must be referenced, because after all, by the First Law, Golden Probability is the fundamental measure of human observation, at least on this scale. A brief breakdown of this proposal works as follows: the Golden Probability of a world without rain is low, and so the rain, relative to human observation, is incredibly likely to exist. Thus it optimized. Then, the Golden Probability of a world where grass takes no aid in rain is low, and so the grass, relative to human observation, is incredibly likely to take aid from rain. Thus it evolved.

Specifically for that second statement, the Outlook Probability of grass thriving without rain is low, and so by the Second Law, the Golden Probability of this is also low. And even if grass holds higher Golden Linkage than a simple zero, the Golden Probability can only go lower from there. Thus, the Golden Probability of a rainless grass existing is low. Consequently, grass evolved to take aid from rain. Causality, now, may become more complicated than originally meets the eye in the simplistic world of pure Outlook Probability. In the world of Golden Probability, optimization takes the spotlight. But technically, in this particular viewpoint-based world, are such things *incorrect*? No, it seems. By the function of human birth described in the First Law, technically this form of causality, within its scope, is no less correct and no less relevant than the “standard” form of causality.

Also note that one may very feasibly argue that this identity of causality between rain and grass is only relative to the Golden Point, and that while technically human life is a very good approximation of such things under the right conditions, per the Golden Principle, it still is technically a different subject. For this situation, then, the resolution is simple: use of the trace connection. We would rewrite the statement as follows: *the rain optimized to enable the Golden Point, human observation trace connected to the Golden Point, and some particular grass evolved*

to take advantage of the optimization of rain.

19 Valuable Equations

Some of the most valuable equations and formulations behind Auruspectation will be displayed in this section.

A great place to start regards the probability of the Golden Point before and after a mission holding a Golden Linkage of L . The likelihood of the Golden Point directly before the mission is some $pa + (1 - p)b$, where p is the Outlook Probability of the mission succeeding, with a and b being the likelihoods of the Golden Point if the mission does and does not happen, respectively. Let this value be represented as follows.

$$\varphi_0 = pa + (1 - p)b$$

Comparatively, if the mission succeeds, then the Golden Point probability is simply a , or representatively, the content below.

$$\Phi_0 = a$$

And because that mission has a Golden Linkage of L , this means that $b = a \times \frac{1-L}{1+L}$, which may be substituted into the full formulatio.

$$\varphi_0 = pa + (1 - p)a \times \frac{1 - L}{1 + L}$$

With this considered, both as may be swapped out for Φ_0s , which may be configured and factored.

$$\frac{\varphi_0}{\Phi_0} = p + \frac{1 - L}{1 + L} (1 - p)$$

Last but not least, it is helpful to imagine what an observer would note about this situation in practice. It can safely be assumed that periods of Goodness decline are rare, because as was seen from the Golden Trek-of-Nescecoins framework, even gaining just one Nescecoin with time approximately halves the Golden Probability of the entire Golden Trek. This means that the Goodness in the times leading up to the mission, and then following it, will be overwhelmed

by Goodness increase, making declining net Goodness thereabout a near impossibility. All things considered, actual Goodness values taken should consistently be less than φ_0 if before, and greater than Φ_0 if after, due to the effects of this perpetually rising Goodness. This means that if φ_0 is the direct value before, then $\varphi < \varphi_0$, and likewise $\Phi > \Phi_0$ for the value directly after. Hence, $\frac{\varphi}{\Phi} < \frac{\varphi_0}{\Phi_0}$. Therefore an algebraic combination may be performed.

$$\frac{\varphi}{\Phi} < p + \frac{1-L}{1+L} (1-p)$$

Note that there is also a form to keep L as the primary arbiter of inequalities.

$$L < \frac{\Phi - \varphi}{\Phi + \varphi - 2\Phi p}$$

And of course there is also the Difficulty Dynamics approach that can be taken, removing Outlook Probability from the system in its entirety.

$$\frac{\varphi}{\Phi} < 1 - \frac{2L}{(1+L)(1+\xi)}$$

This all being said, perhaps in many cases a more useful quantity to use in place of Golden Linkage is Golden Probability, written with g . In this case, due to the nature of the conversion, the inequality becomes a remarkably simple final result.

$$\frac{\varphi}{\Phi} < \frac{p}{g}$$

One example of this in action regards Optimal Missions. Such missions can be defined to have a Golden Probability following $g > \frac{1}{2}$, and thus in their case, they must have a ratio from the Goodness before to after which is less than double the Outlook Probability of mission success.

$$\frac{\varphi}{\Phi} < 2p$$

This simply means that for all Optimal Missions, (which by the Third Law, is expected to be any observed mission which is not outwardly insignificant), $\frac{\varphi}{\Phi} < 2p$ will hold. For any Optimal Mission, the ratio of some Goodness predating it to some Goodness

postdating it shall be less than twice the probability of that Optimal Mission occurring successfully. For Prerequisites, of course, this should be even more extreme of a compression of possibilities, because they serve as a subclass to Optimal Missions, but with only additional importance of occurrence to the Golden Mission. In their case, $g \approx 1$, which yields $\frac{\varphi}{\Phi} < p$. Note that even if the Golden Probability is ever so slightly less than one, as should be the case for the world of finite Nescecoins in the corresponding Imperfection Magnitude Drop, there are always bound to be extra properties of the physical Earthen event not counted in the Prerequisite's definition, and if these effects even slightly nudge down the Goodness of the before case relative to the after case, a feat which they should easily do, then $\frac{\varphi}{\Phi} < p$ will be bound to hold brilliantly. Thus, where any Optimal Mission will follow $\frac{\varphi}{\Phi} < 2p$, any Prerequisite will follow a stricter bound.

$$\frac{\varphi}{\Phi} < p$$

One interesting Nescecoin-based statistic involves failed missions in the Golden Trek. If the mission itself has a Golden Linkage of L , then the extra Nescecoins picked up, at least with respect to if it had succeeded, is precisely the following.

$$\log_2 \left(\frac{1+L}{1-L} \right)$$

20 Tournament Distribution

For a perfect Nescecoin based Golden Trek, each Nescecoin gained with time halves the preexisting Golden Probability of a certain Golden Trek. But life, as similar as it may be to this framework in certain respects, is also much more complicated than this. Perhaps, then, an alternative - not necessarily better or worse - just different - is for a simple tournament to be used as a framework instead. For this tournament, an integer power of two contestants are entered to align with a corresponding integer of rounds. Each contestant is given a numerical label, ranging from one to the total number of contestants, inclusive, such that every number is used exactly once. Then, these contestants are placed into random starting locations,

and the tournament is initiated. Each time a game in the tournament is run, to decide who will move on to the next round, the probability of some contestant with numerical label a defeating their opponent, with numerical label b is precisely $\frac{a}{a+b}$. From here, the entirety of the tournament is run, until a single winner emerges. Let the model being “successful” exactly correspond to player one winning, the least likely winner out of everyone, such that such a win corresponds to the Golden Point under the Rare Earth Hypothesis. Any other winner of the tournament is counted as a “loss” overall, corresponding to the Departure Point.

The most obvious characteristic of the tournament is the total likelihood of it being “successful”, with the winning of player one, itself being highly unlikely for several rounds. But perhaps another key aspect of the tournament is the analogy of “Golden Probability” of certain arrangements of the tournament, such as starting positions and outcomes throughout. If the total probability of player one winning is written as G , then the Golden Probability may be written as $g = \frac{p}{G}$, where p is the Outlook Probability of such an arrangement coming to pass. This is a generalization of the binary form of $g = \frac{p a}{p a + (1-p) b}$, such that all possibilities are taken into account rather than just two, making $g = \frac{p a}{G}$, and also such that the Golden Point is a guarantee for all analyzed arrangements, making $a = 1$. Thus, $g = \frac{p}{G}$. Therefore, the Golden Probability can be taken as proportional to the Outlook Probability of a certain arrangement and configuration occurring, with $g \propto p$, or $g = k p$. The Outlook Probability itself is straightforward to calculate, by multiplying together the probability of each and every round playing out as it did, with $\frac{a}{a+b}$, where a is the winner and b is the loser. Thus, the Golden Probability can be taken to be proportional to this total product.

From here, the question becomes, what is the probability distribution of each Golden Probability across the set of all possible scenarios in which player one wins?

I ran this exact algorithm in a computer simulation over several days, eventually mapping out hundreds of points onto the cumulative distribution function

corresponding to the answer to this question. Amazingly, it came out to the following.

$$F_X(x) = \text{li}^{-1}(k x)$$

This is expressed such that $\text{li}(x)$ is the logarithmic integral function from $x = 0$ to $x = 1$, which itself is bijective, such that the inverse function, $\text{li}^{-1}(x)$, may exist. If the Golden Probability x is scaled by the proper coefficient k , (a strictly negative value) and then run through such a function, the percentile on the scale from 0 to 1 may be derived, with $F_X(x) = \text{li}^{-1}(k x)$. As for the probability density function corresponding to this concept, it may be taken as follows.

$$\frac{d}{dx} F_X(x) = \frac{d}{dx} \text{li}^{-1}(k x)$$

Assuming momentarily that $k = 1$, then this is to be the case.

$$f_X(x) = \frac{1}{\text{li}'(\text{li}^{-1}(x))}$$

Two identities are needed to further simplify this query.

$$\text{li}(e^x) = \text{Ei}(x)$$

$$\text{li}(x) = \int_0^x \frac{dt}{\ln(t)}$$

From here, it can be seen that $f_X(x) = \ln(\text{li}^{-1}(x))$. Rearranging from here yields another identity.

$$x = \text{li}(e^{f_X(x)})$$

Due to the identity of $\text{li}(e^x) = \text{Ei}(x)$, substitution allows for the next step.

$$x = \text{Ei}(f_X(x))$$

Thus, the derivative is found.

$$f_X(x) = \text{Ei}^{-1}(x)$$

As for reinstating the constant of proportionality, that can be done using substitution and the chain rule.

$$f_X(x) = k \text{Ei}^{-1}(k x)$$

One further interesting aspect of the probability density function of the tournament distribution,

is that it enables the calculation of the expected value³². This would simply be the result of the following value, which may be written with an unknown $\mathbb{E}$.

$$\mathbb{E} = \frac{\int_0^\infty k \operatorname{Ei}^{-1}(kx) x dx}{\int_0^\infty k \operatorname{Ei}^{-1}(kx) dx}$$

The k coefficients may easily be factored out, or perhaps replaced with values of -1 to ensure negatives are applied in the right places, and the remaining k values, (found within the inverse exponential integral functions), may be replaced with -1 s as well, because they are merely scale factors of the arbitrary multiplicative arrangement of the infinitesimal precise Golden Probabilities, and thus have no real bearing on the percentile scale, which we are after. -1 is the simplest of the negative numbers, so therefore it will make for the most efficient substitution upfront. If x itself is then replaced with its opposite, and the signs on the integrals are updated accordingly, which by the commutative property will not affect the net sums, then the integral may be rewritten in a cleaner form.

$$\mathbb{E} = \frac{\int_{-\infty}^0 \operatorname{Ei}^{-1}(x) x dx}{\int_{-\infty}^0 \operatorname{Ei}^{-1}(x) dx}$$

Due to conservation of area, it follows that $\int_{-\infty}^0 \operatorname{Ei}^{-1}(x) dx = \int_{-\infty}^0 \operatorname{Ei}(x) dx$, which itself is simply equal to a unit negative one. Therefore the entire formulation is $\mathbb{E} = -\int_{-\infty}^0 \operatorname{Ei}^{-1}(x) x dx$. The indefinite integral itself of $\int \operatorname{Ei}^{-1}(x) x dx$ may be solved through integration by parts, which gives it an equivalent form as follows, (taking advantage of the identity that $\frac{d}{dx} \operatorname{li}^{-1}(x) = \operatorname{Ei}^{-1}(x)$).

$$\int \operatorname{Ei}^{-1}(x) x dx = \operatorname{li}^{-1}(x) x - \int \operatorname{li}^{-1}(x) dx$$

³²Note that since it is already apparent that the scale factor k upon things is arbitrary, and all that it truly mathematically important is the relative ratios of the returned Golden Probabilities within this function, as they are infinitesimal anyway, we can thus conclude that the actual percentile / relative linear ranking of the expected tournament / Golden Trek or what have you which satisfies the winning of player one, given by $F_X(\mathbb{E})$, should be the most fundamental result to process. But to start, determining the $\mathbb{E}$ itself will be helpful.

Next, the $\int \operatorname{li}^{-1}(x) dx$ itself may be expanded.

$$\int \operatorname{li}^{-1}(x) dx = \operatorname{Ei}(2 \ln(\operatorname{li}^{-1}(x))) + C$$

(Note the additive cancellation of two instances of $x \operatorname{li}^{-1}(x)$ which would have been present in the expansion by according to the integral-of-inverse-function formula in combination with the integral of the non-inverted logarithmic integral function). Notice that due to the identity transforming the exponential integral into the logarithmic integral with an in-function application of the exponential function e^x , this has a better representation still.

$$\int \operatorname{li}^{-1}(x) dx = \operatorname{li}(\operatorname{li}^{-1}(x)^2) + C$$

Reconfiguring this into the original formulation yields the net result of that indefinite integral.

$$\int \operatorname{Ei}^{-1}(x) x dx = \operatorname{li}^{-1}(x) x - \operatorname{li}(\operatorname{li}^{-1}(x)^2) + C$$

Due to the fact that this indefinite integral is zero when $x = 0$ and $C = 0$, it follows that the desired result of $\mathbb{E} = -\int_{-\infty}^0 \operatorname{Ei}^{-1}(x) x dx$ is in fact the following limit.

$$\mathbb{E} = \lim_{x \rightarrow -\infty} \operatorname{li}^{-1}(x) x - \operatorname{li}(\operatorname{li}^{-1}(x)^2)$$

If some t is taken such that $t = \operatorname{li}^{-1}(x)$, then the limit is equivalent to the following.

$$\mathbb{E} = \lim_{t \rightarrow 1} \operatorname{li}(t^2) - t \operatorname{li}(t)$$

Thus, $\mathbb{E}$ may officially be found.

$$\mathbb{E} = \ln(2)$$

Note that this expected value, $\mathbb{E}$, scales in proportion depending on that original k scale factor, which was disregarded in the making of it. That is, $\mathbb{E} = \ln(2)$ only when $k = -1$, otherwise it scales inversely proportionally with k . Therefore, perhaps the most fundamental representation of this expectancy is its corresponding “percentile”, so to speak, in the scope of all possible tournaments / Golden Treks which may

be associated with the win of player one. This ranking value, given by $F_X(\mathbb{E})$, is more fundamental, because it does not change with k , and instead is fixed. Low and behold, this “expected ranking” of sorts for the tournament distribution, is given by the following.

$$\text{li}^{-1}(-\ln(2)) \approx \frac{2}{3}$$

That is, in the set of all possible circumstances which could have led to player one winning the tournament, the statistical expectancy is for the Golden Probability analog of the particular set of events which transpired to be greater than two thirds of all possible outcomes, but less than one third of them, placing that expectancy at just about the 67th percentile overall. This, perhaps, has at least a rough correlation to our own Golden Trek’s happenings.

Knowing this, one can relocate the percentile in the set of all Golden Points that would ensue if a Golden Point picked up a Nescecoin. In general, if the probability of the Golden Point were altered by some ratio s , then according to this tournament framework, if the original percentile were x , then the Golden Probability corresponding to that exact original configuration would be $F_X^{-1}(x) = \frac{\text{li}(x)}{k}$, making the scale factor $\frac{s \text{li}(x)}{k}$. Then, with the new application of the cumulative distribution function, the final new percentile would be this.

$$F_X\left(\frac{s \text{li}(x)}{k}\right) = \text{li}^{-1}(s \text{li}(x))$$

In the case of Nescecoins, with $s = 2^{-N}$, this would become $\text{li}^{-1}\left(\frac{\text{li}(x)}{2^N}\right)$, but the underlying logic is the same. This formulation of $\text{li}^{-1}(s \text{li}(x))$ is quite intimidating at a glance, however viewing it through the form of a differential equation is surprisingly easier to swallow. If $y = \text{li}^{-1}(s \text{li}(x))$, then $\text{li}(y) = s \text{li}(x)$, which yields an implicit derivative result of interest.

$$\frac{dy}{dx} = s \log_x(y) = \frac{s \log(y)}{\log(x)}$$

In terms of Nescecoins again, this translates to $\frac{dy}{dx} = \frac{\log_x(y)}{2^N}$. Also note that these differential equations are only valid if and only if $y = 0$ when $x = 0$, such that

their graph passes through the origin. If that is accepted, then these differential equations, along with the proper application of Euler’s Method or what have you, will return a function which applies a transformation to any Golden Trek percentile of optimality given an additional probability component of s . For reference, with the basic Nescecoin built Golden Trek framework, an application of s additional probability resulted in a percentile factor of s , so that differential equation would look like $\frac{dy}{dx} = s$, by comparison. Now, with this tournament analysis, it is supposed that a better approximation might be $\frac{dy}{dx} = s \log_x(y)$.

Through the solving of $y' = s \log_x(x y')$, as x tends to 0, it can be shown that the slope of the correct curve as the graph passes through the origin should be precisely s itself. Thus, given a Golden Trek success percentile of x , and an additional numerical probabilistic requirement for the Golden Point of s , in its application yielding a new percentile y , the derivative of y with respect to x as both tend to zero is s itself. And because this curve must pass through the origin, this derivative toward zero is itself tending toward the ratio between the cases.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y(x,s)}{x} = s$$

Practically speaking, where $x \approx 0$ and $y \approx 0$, $\frac{y(x,s)}{x} \approx s$. For example, given two Imperfection Magnitude Drops, the ratio of their precise Golden Probabilities of occurrence in the full probability spectrum is nearly equivalent to their ratio in percentiles.

If one single Nescecoin is added to the Golden Trek, then by this logic, the slope of the graph at $x = y = 0$ is simply $\frac{1}{2}$, and then as the function increases from there, the curve begins to overtake the simple $y = \frac{1}{2}x$ in a concave-up fashion, eventually reaching a vertical asymptote at $x = 1$, where the y stretches up towards positive infinity.

21 Onward

Now it is time to look toward the future. If the contents of this paper are even partially valid, then this general field has the potential to be crucial to humanity’s future. It may even become the next layer

of physics according to our descendents, valued just as much as any cosmological equation or law. After all, humanity must always view the scene of reality through the window of the human mind, and in fact we are restricted to this, so whatever general principle we one day attribute to “The Theory of Everything”, must in some way spawn from the human mind’s outlook on the universe rather than simply a state of the universe itself. Thus, it seems logical that the fundamental birthing function of that human mind, as portrayed by Golden Science and its like, may in fact one day be helpful to the pursuit of the Theory of Everything, or at least some future iteration of it could be. Specifically, a key question might be to answer why we were collectively born in such a low entropy universe, and additionally, why we were born in such extreme proximity to a Golden Point in this low entropy universe. The two seem to be asking almost the same question, so if they could be collectively solved, such a solution might be key to explaining reality as a whole. Hence, the Theory of Everything could be progressed from this realization. Furthermore, helping to uncover the state of events which led us to this modern day might also be helpful. Questions may be posed such as “is mammalian history personal history, just personal history forgotten?” or “what function determines our birth place in the set of all reality?”. If these could be answered thoroughly, then the applications could be extraordinary.

In the scope of all this new application, beginning even now with what we currently know, is how we might choose to process this wealth of information regarding the Rare Earth Hypothesis itself. If any of the conclusions by the Rare Earth Hypothesis or even this paper are valid, then it seems necessary that our language should adapt accordingly. Typically, but not as a rule, it seems as though Great Filters in the past should be regarded in the positive connotation. Unless one disagrees about the beauty of human life and the Golden Point, it would be a natural conclusion that the Great Filters which enabled such things, and crucially, worked splendidly, should be regarded positively. Thus, I propose that the positive connotation of a Great Filter should be the word “Miracle”, spelled specifically with a cap-

ital “M”, to separate it from everyday miracles. If the event is not explicitly being regarded positively, then the simple Great Filter will do. But it seems that we as a society should be very thankful to these events of sheer success which against all odds enabled our existence, so in any of the contexts where they are regarded with praise, the word Miracle should be used, both in the scientific discourse community and the general public. The logic behind our place in the universe, and even the Fermi Paradox, is often viewed through a very cold and unfeeling light. If the Golden Principle were valid, then, it would be imperative to flip this mindset on its head, and instead teach future generations about the luck and absolute beauty of our world. The Miracles, then, could be the foundational centerpiece of this new image. That is, as the science progresses into the unforeseen mists of Golden Science and its like, it is crucial that the societal image of this progress be lit with a sense of vibrance and awe to accurately reflect on how astonishing it all really is. This is a message to all parents, teachers, and young people reading this. So long as this paper is not entirely mistaken, each and every one of you is some form of special. We cannot yet understand why, but we all live just before the most foundational moment in the history of life. It is the payoff, so to speak. So it is your honor and your duty to protect the Earth, protect knowledge itself, and keep asking questions. Everything you see around you is a marvel in the cosmos orders of magnitude beyond any fathomability. And the center of this marvel is you, here and now, present with the very choices and wonder you are ready to create.

22 Glossary

Attribute A characteristic of a Golden Trek, measured by probability of occurrence.

Attribute Set A set of one or more Attributes.

Auruman An extension of the human species including all human ancestral organisms as part of the group.

Auruspective The visualization of the past from

the lens of Auruman experience.

Departure Point The temporal point after which a biosphere becomes too incapable at self preservation to progress past that point, permanently overriding the chances of a Golden Point.

Earth Route A Route having directly to do with Earth's Golden Trek.

Empty Fluke A Fluke with near-zero Golden Linkage.

Fluke A Mission which holds a very low Outlook Probability of occurrence.

Golden Linkage The measure of dependency of the Golden Point on an Attribute.

Golden Mission A specific type of Mission for which the Attribute being changed is the absence of a Golden Point to an instance of the Golden Point.

Golden Point The temporal point after which a biosphere becomes too capable in self preservation to regress past that point.

Golden Probability The probability of an Attribute in the set of all scenarios corresponding to a Golden Point.

Golden Trek The story of all events and factors which lead to a Golden Point

Goodness The Outlook Probability of the Golden Point occurring.

Imperfection Magnitude Drop The term for when a Mission within a Golden Trek according with an Attribute Set is vastly lowered in Optimality with respect to the same Mission in the scope of an Inverse Attribute Set, and alternatively can be named to an Attribute in the Attribute Set which actually caused this significant Optimality lowering.

Inverse Attribute An Attribute which, relative to a given Attribute, is taken as present if and only if said given Attribute is absent.

Inverse Attribute Set An Attribute Set for which all Attribute elements are Inverse Attributes with respect to a specified Attribute Set.

L-Threshold The lowest possible value of Golden Linkage for a non-outwardly-insignificant Attribute of Earth's Golden Trek by the Third Law of Auruspectation.

Miracle A Prerequisite that is also a Fluke.

Mission A temporal change in a temporally based Attribute.

Nonoptimal Mission A Mission which is not an Optimal Mission.

Outlook Probability The probability of an Attribute in the set of all possibilities, Golden Point corresponding or not.

Optimal Mission A Mission whose success would be more Optimal to the Golden Trek than its failure. (*Golden Probability* $> \frac{1}{2}$).

Optimality The measure of placement in the broad Optimal Percentile Scale of a Route amongst all possible conceivable Routes, as pertaining to a desired Mission.

Prerequisite A Mission which, if failed, would result directly in an Imperfection Magnitude Drop of the Golden Mission.

Route An Attribute Set containing an infinite string of all transpired Attributes over the course of a Mission, altogether being measured with infinitesimal probability of occurrence.

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